

MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Fundamental of Algebra

(English Medium)

Copyright Statement

All rights including trademark and copyrights and rights of translation etc. reserved and vested exclusively with ALLEN Career Institute Private Limited. (ALLEN)

No part of this work may be copied, reproduced, adapted, abridged or translated, transcribed, transmitted, stored or distributed in any form retrieval system, computer system, photographic or other system or transmitted in any form or by any means whether electronic, magnetic, chemical or manual, mechanical, digital, optical, photocopying, recording or otherwise, or stored in any retrieval system of any nature without the written permission of the Allen Career Institute Private Limited. Any breach will entail legal action and prosecution without further notice.

This work is sold/distributed by Allen Career Institute Private Limited subject to the condition and undertaking given by the student that all proprietary rights (under the Trademark Act, 1999 and Copyright Act, 1957) of the work shall be exclusively belong to ALLEN Career Institute Private Limited. Neither the Study Materials and/or Test Series and/or the contents nor any part thereof i.e. work shall be reproduced, modify, re-publish, sub-license, upload on website, broadcast, post, transmit, disseminate, distribute, sell in market, stored in a retrieval system or transmitted in any form or by any means for reproducing or making multiple copies of it.

Any person who does any unauthorised act in relation to this work may be liable to criminal prosecution and civil claims for damages. Any violation or infringement of the propriety rights of Allen shall be punishable under Section- 29 & 52 of the Trademark Act, 1999 and under Section- 51, 58 & 63 of the Copyright Act, 1957 and any other Act applicable in India. All disputes are subjected to the exclusive jurisdiction of courts, tribunals and forums at Kota, Rajasthan only.

Note:- This publication is meant for educational and learning purposes. All reasonable care and diligence have been taken while editing and printing this publication. ALLEN Career Institute Private Limited shall not hold any responsibility for any error that may have inadvertently crept in. ALLEN Career Institute Private Limited is not responsible for the consequences of any action taken on the basis of this publication.

FUNDAMENTALS OF ALGEBRA

1. INDICES AND SURDS

1.1 Indices

Definition of Indices :

If 'a' is any non zero real or imaginary number and 'm' is a positive integer, then $a^m = a \cdot a \cdot a \dots a$ (m times). Here a is called the base and m is the index, power or exponent.

Law of indices :

- (i) $a^0 = 1, (a \neq 0)$
- (ii) $a^{-m} = \frac{1}{a^m}, (a \neq 0)$
- (iii) $a^m \cdot a^n = a^{m+n}$
- (iv) $\frac{a^m}{a^n} = a^{m-n}, a \neq 0$
- (v) $(a^m)^n = a^{mn} = (a^n)^m$
- (vi) $\sqrt[q]{a^p} = a^{p/q}, q \in \mathbb{N} \text{ and } q \geq 2$
- (vii) If $x = y$, then $a^x = a^y$, but the converse may not be true. e.g. : $(1)^6 = (1)^8$, but $6 \neq 8$

For $a^x = a^y$ we have following possibilities

- If $a \neq \pm 1, 0$, then $x = y$
- If $a = 1$, then x, y may be any real number
- If $a = -1$, then x, y may be both even or both odd
- If $a = 0$, then x, y may be any positive real number

But if we have to solve the equations like $(f(x))^{g(x)} = (f(x))^{h(x)}$ (i.e same base, different indices) then we have to solve :

- (a) $f(x) = 1$ (b) $f(x) = -1$ (c) $f(x) = 0$ (d) $g(x) = h(x)$

Verification should be done in (b) and (c) cases

Illustration-1 : Solve $(4-x)^{x^3-4x} = (4-x)^{(x^2-4)(x-1)}$

Solution : **Case-1 :** $4-x = 1 \Rightarrow x = 3$

Case-2 : $4-x = -1 \Rightarrow x = 5$

for $x = 5$, $x^3 - 4x$ is odd

and $(x^2 - 4)(x - 1)$ is even

so $x = 5$ does not satisfy the given equation

Case-3 : $4-x = 0 \Rightarrow x = 4$

For $x = 4$, $x^3 - 4x > 0$ and $(x^2 - 4)(x - 1) > 0$

$\Rightarrow x = 4$ satisfies the given equation

Case-4 : $x^3 - 4x = (x^2 - 4)(x - 1)$

$\Rightarrow x^3 - 4x = x^3 - x^2 - 4x + 4$

$\Rightarrow x^2 - 4 = 0, \Rightarrow x = \pm 2$ which satisfies the given equation.

From case-1, case-2, case-3 and case-4, solutions set of the given equation is $\{-2, 2, 3, 4\}$

(viii) If $a^x = b^x$ then consider the following cases :

- If $a \neq \pm b$, then $x = 0$
- If $a = b \neq 0$, then x may have any real value for which a^x is well defined.
- If $a = -b \neq 0$, then x is even.
- If $a = b = 0$, then x can be any positive real.

If we have to solve the equation of the form $[f(x)]^{h(x)} = [g(x)]^{h(x)}$, i.e., same index, different bases, then we have to solve :

(a) $f(x) = g(x)$ (b) $f(x) = -g(x)$ (c) $h(x) = 0$

Verification should be done in (a), (b) and (c) cases.

Illustration-2: Solve $(x^2 - 4)^{2x} = (x^2 + 2x)^{2x}$.

Solution : **Case-1 :** when $x = 0$

Base : $x^2 + 2x = 0$, so $x = 0$ does not satisfy given equation

Case-2 : $x^2 - 4 = x^2 + 2x \neq 0$

$\Rightarrow x \in \phi$

Case-3 : $x^2 - 4 = -x^2 - 2x \neq 0 \Rightarrow 2x^2 + 2x - 4 = 0$

$x^2 + x - 2 = 0 \Rightarrow x = 1$

satisfies the given equation

Case-4 : $x^2 - 4 = x^2 + 2x \Rightarrow x = -2$

$x = -2$ does not satisfy the given equation.

From all above cases, $x \in \{1\}$

Illustration-3: If $a^x = b$, $b^y = c$, $c^z = a$, prove that $xyz = 1$, where a, b, c are distinct numbers.

Solution : We have, $a^{xyz} = (a^x)^{yz}$

$\Rightarrow a^{xyz} = (b)^{yz} [\because a^x = b]$

$\Rightarrow a^{xyz} = (b^y)^z$

$\Rightarrow a^{xyz} = c^z [\because b^y = c]$

$\Rightarrow a^{xyz} = a [\because c^z = a]$

$\therefore a^{xyz} = a^1$

$\Rightarrow xyz = 1$

1.2 Surds

If 'x' is a rational number, which is not the n^{th} power ($n \in \mathbb{N} \setminus \{1\}$) of any rational number, then the number $x^{1/n}$ usually denoted by $\sqrt[n]{x}$ is called surd. The sign ' $\sqrt[n]{}$ ' is called the radical sign. The number in the angular part of the sign, i.e., 'n' is called order of the surd. In case of $n = 2$ the expression $\sqrt[n]{x}$, simply written as $\sqrt{x}$.

Note :

- If $\sqrt[n]{x}$ is a surd then $-(\sqrt[n]{x})$ is also a surd.
- Every surd is an irrational number (but every irrational number is not a surd).
- To rationalize the denominator of a fraction of the form $\frac{a}{\sqrt{b}}$, multiply the numerator and

denominator of the fraction by $\sqrt{b} \Rightarrow \frac{a}{\sqrt{b}} = \frac{a}{\sqrt{b}} \cdot \frac{\sqrt{b}}{\sqrt{b}} = \frac{a\sqrt{b}}{\sqrt{b^2}} = \frac{a\sqrt{b}}{b}$.

Eg.

(a) $\sqrt{3}$ is a surd and $\sqrt{3}$ is an irrational number.

(b) $\sqrt[3]{5}$ is surd and $\sqrt[3]{5}$ is an irrational number.

(c) π is an irrational number, but it is not a surd.

1.2.1 Conjugate of a Surd

If two binomial surds (surd containing two terms such as $2 + \sqrt{3}$, $2\sqrt{5} - \sqrt{7}$ etc) are such that only the sign connecting the individual terms are different, then they are said to be conjugate of each other. If these surds are quadratic, then their product would always be rational. So in case of a binomial quadratic surd, we use its conjugate as its rationalizing factor.

Ex. Conjugate of $3\sqrt{2} + \sqrt{5}$ is $3\sqrt{2} - \sqrt{5}$ or $-3\sqrt{2} + \sqrt{5}$

Illustration-4 : Rationalize the denominator of $\frac{1}{3\sqrt{2} + \sqrt{5}}$

Solution :

A conjugate of $3\sqrt{2} + \sqrt{5}$ is $3\sqrt{2} - \sqrt{5}$

Therefore multiplying the conjugate in the numerator and denominator of the given fraction.

$$\frac{3\sqrt{2} - \sqrt{5}}{(3\sqrt{2} + \sqrt{5})(3\sqrt{2} - \sqrt{5})}$$

$$= \frac{3\sqrt{2} - \sqrt{5}}{(3\sqrt{2})^2 - (\sqrt{5})^2}$$

$$= \frac{3\sqrt{2} - \sqrt{5}}{18 - 5}$$

$$= \frac{3\sqrt{2} - \sqrt{5}}{13}$$

Illustration-5 : Using the fact that $(\sqrt{x} + \sqrt{y})^2 = x + y + \sqrt{xy}$ to find the square root of $7 + 2\sqrt{10}$. If this

number can be expressed in the form $\sqrt{a} + \sqrt{b}$, where $a \leq b$, find the value of $b - a$.

(A) 3

(B) 4

(C) 0

(D) 2

Solution :

Let $\sqrt{7 + 2\sqrt{10}} = \sqrt{a} + \sqrt{b}$, where $a, b \in \mathbb{Q}$

$$\Rightarrow 7 + 2\sqrt{10} = a + b + 2\sqrt{ab}$$

$$\Rightarrow a + b = 7 \text{ and } \sqrt{ab} = \sqrt{10}$$

$$(b - a)^2 = (a + b)^2 - 4ab$$

$$= 49 - 40 = 9$$

$$\Rightarrow \sqrt{(b - a)^2} = 3 \Rightarrow b - a = 3 \text{ (since } b > a)$$

OR

$$7 + 2\sqrt{10} = 7 + 2 \cdot \sqrt{2} \cdot \sqrt{5} = (\sqrt{2})^2 + (\sqrt{5})^2 + 2 \cdot \sqrt{2} \cdot \sqrt{5} = (\sqrt{2} + \sqrt{5})^2$$

$$\text{So } \sqrt{7 + 2\sqrt{10}} = \sqrt{2} + \sqrt{5} \therefore b - a = 3$$

Illustration-6 : If $x = \frac{1}{2 + \sqrt{3}}$, find the value of $x^3 - x^2 - 11x + 4$.

Solution : as $x = \frac{1}{2 + \sqrt{3}} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}} = \frac{2 - \sqrt{3}}{(2)^2 - (\sqrt{3})^2}$

$$x = \frac{2 - \sqrt{3}}{4 - 3} = 2 - \sqrt{3}$$

$$x - 2 = -\sqrt{3}, \text{ squaring both sides, we get}$$

$$(x-2)^2 = (-\sqrt{3})^2 \Rightarrow x^2 + 4 - 4x = 3$$

$$\Rightarrow x^2 - 4x + 1 = 0$$

$$\text{Now } x^3 - x^2 - 11x + 4$$

$$= x^3 - 4x^2 + x + 3x^2 - 12x + 4$$

$$= x(x^2 - 4x + 1) + 3(x^2 - 4x + 1) + 1 = x \times 0 + 3(0) + 1 = 0 + 0 + 1 = 1$$

Illustration-7 : If $x = 3 - 2\sqrt{2}$, find $x^2 + \frac{1}{x^2}$

Solution : We have, $x = 3 - 2\sqrt{2}$.

$$\therefore \frac{1}{x} = \frac{1}{3 - 2\sqrt{2}} = \frac{1}{3 - 2\sqrt{2}} \times \frac{3 + 2\sqrt{2}}{3 + 2\sqrt{2}}$$

$$= \frac{3 + 2\sqrt{2}}{(3)^2 - (2\sqrt{2})^2} = \frac{3 + 2\sqrt{2}}{9 - 8} = 3 + 2\sqrt{2}$$

$$\text{Thus, } x^2 + \frac{1}{x^2} = (3 - 2\sqrt{2})^2 + (3 + 2\sqrt{2})^2$$

$$= 2((3)^2 + (2\sqrt{2})^2) = 2(9 + 8) = 34$$

Illustration-8 : Rationalise the denominator of $\frac{1}{\sqrt{3} - \sqrt{2} - 1}$

Solution : $\frac{1}{\sqrt{3} - \sqrt{2} - 1} = \frac{1}{\sqrt{3} - \sqrt{2} - 1} \times \frac{\sqrt{3} + \sqrt{2} + 1}{\sqrt{3} + \sqrt{2} + 1}$

$$= \frac{\sqrt{3} + \sqrt{2} + 1}{(\sqrt{3} - \sqrt{2} - 1)(\sqrt{3} + \sqrt{2} + 1)} = \frac{\sqrt{3} + \sqrt{2} + 1}{(\sqrt{3})^2 - (\sqrt{2} + 1)^2} = \frac{\sqrt{3} + \sqrt{2} + 1}{-2\sqrt{2}} = -\left(\frac{\sqrt{6} + \sqrt{2} + 2}{4}\right)$$

Illustration-9 : Evaluate the following

(i) $(\sqrt[3]{64})^{\frac{-1}{2}}$ (ii) $\left(\frac{121}{169}\right)^{-3/2}$

Solution : (i) $(\sqrt[3]{64})^{\frac{-1}{2}} = \left[(64)^{\frac{1}{3}}\right]^{\frac{-1}{2}} = (64)^{\frac{1}{3} \times \frac{-1}{2}} = (64)^{\frac{-1}{6}}$

$$= (2^6)^{\frac{-1}{6}} = 2^{6 \times \left(\frac{-1}{6}\right)} = 2^{-1} = \frac{1}{2}$$

(ii) $\left(\frac{11 \times 11}{13 \times 13}\right)^{-3/2} = \left(\frac{11^2}{13^2}\right)^{-3/2} = \left(\frac{11}{13}\right)^{2 \times \frac{-3}{2}} = \left(\frac{11}{13}\right)^{-3} = \left(\frac{13}{11}\right)^3 = \frac{13^3}{11^3} = \frac{2197}{1331}$

Illustration 10 : Simplify $\left[\sqrt[3]{\sqrt[6]{a^9}}\right]^4 \left[\sqrt[6]{\sqrt[3]{a^9}}\right]^4$

(A) a^{16} (B) a^{12} (C) a^8 (D) a^4

Solution. $a^{9(1/6)(1/3)4} \cdot a^{9(1/3)(1/6)4} = a^2 \cdot a^2 = a^4.$

Illustration 11 : Simplify $a \left(\frac{\sqrt{a} + \sqrt{b}}{2b\sqrt{a}}\right)^{-1} + b \left(\frac{\sqrt{a} + \sqrt{b}}{2a\sqrt{b}}\right)^{-1}$

Solution. The given expression is equal to

$$a \left(\frac{2b\sqrt{a}}{\sqrt{a} + \sqrt{b}}\right) + b \left(\frac{2a\sqrt{b}}{\sqrt{a} + \sqrt{b}}\right)$$

$$= \frac{2ab}{\sqrt{a} + \sqrt{b}} (\sqrt{a} + \sqrt{b}) = 2ab$$

Illustration 12 : Evaluate $\sqrt{3 + \sqrt{3} + \sqrt{2 + \sqrt{3} + \sqrt{7 + \sqrt{48}}}}$

Solution.

$$\sqrt{3 + \sqrt{3} + \sqrt{2 + \sqrt{3} + \sqrt{7 + \sqrt{48}}}}$$

$$= \sqrt{3 + \sqrt{3} + \sqrt{2 + \sqrt{3} + \sqrt{4 + 3 + 2\sqrt{12}}}} = \sqrt{3 + \sqrt{3} + \sqrt{2 + \sqrt{3} + \sqrt{4} + \sqrt{3}}}$$

$$= \sqrt{3 + \sqrt{3} + \sqrt{3} + 1} = \sqrt{4 + 2\sqrt{3}} = \sqrt{3} + 1 = \sqrt{3} + 1$$

Illustration 13 : Find rational numbers a and b , such that $\frac{4+3\sqrt{5}}{4-3\sqrt{5}} = a + b\sqrt{5}$

Solution.
$$\frac{4+3\sqrt{5}}{4-3\sqrt{5}} \times \frac{4+3\sqrt{5}}{4+3\sqrt{5}} = a + b\sqrt{5}$$

$$\frac{61+24\sqrt{5}}{-29} = a + b\sqrt{5}$$

$$a = -\frac{61}{29}, b = -\frac{24}{29}$$

Do yourself-1 :

Only One Option is Correct for Q.1 to Q.5

1. $\left(\left((625)^{-1/2} \right)^{1/4} \right)^2 =$

- (A) 4 (B) 5 (C) 2 (D) 3

2. $\left(5 \left(8^{1/3} + 27^{1/3} \right)^3 \right)^{1/4} =$

- (A) 3 (B) 6 (C) 5 (D) 4

3. $\left\{ 4 \left(\frac{1}{x} \right)^{-12} \right\}^{-2/3} =$

- (A) $\frac{1}{x^2}$ (B) $\frac{1}{x^4}$ (C) $\frac{1}{x^3}$ (D) $\frac{1}{x}$

4. $\frac{\sqrt{x^3} \times \sqrt[3]{x^5}}{5\sqrt{x^3}} \times \sqrt[30]{x^{77}} =$

- (A) $x^{76/15}$ (B) $x^{78/15}$ (C) $x^{79/15}$ (D) $x^{77/15}$

5. If $\sqrt[4]{3\sqrt{x^2}} = x^k$, then $k =$

- (A) $\frac{2}{6}$ (B) 6 (C) $\frac{1}{6}$ (D) 7

6. Simplify :

(i) $\frac{(5\sqrt{3} + \sqrt{50})(5 - \sqrt{24})}{(\sqrt{75} - 5\sqrt{2})}$

(ii) $\frac{3\sqrt{2}}{\sqrt{6} + \sqrt{3}} - \frac{4\sqrt{3}}{\sqrt{6} + \sqrt{2}} + \frac{\sqrt{6}}{\sqrt{3} + \sqrt{2}}$

$$(iii) \frac{\sqrt{[6+2\sqrt{3}+2\sqrt{2}+2\sqrt{6}]}-1}{\sqrt{5+2\sqrt{6}}}$$

$$(iv) \frac{2 \cdot 3^{n+1} - 7 \cdot 3^{n-1}}{3^{n+1} + 2\left(\frac{1}{3}\right)^{1-n}}$$

$$(v) \left(\frac{1}{3}\right)^{-10} \cdot 27^{-3} + \left(\frac{1}{5}\right)^{-4} \cdot (25)^{-2} + \left(64^{-\frac{1}{9}}\right)^{-3}$$

7. If $\frac{(2^{n+1})^m (2^{2n}) 2^n}{(2^{m+1})^n (2^{2m})} = 1$, then find the value of $\frac{m}{n}$

8. Find rational numbers a and b , such that $\frac{2+3\sqrt{5}}{1-3\sqrt{5}} = a + b\sqrt{5}$

9. The square root of $11 + \sqrt{112}$ is $a + \sqrt{b}$, $a, b \in \mathbb{N}$ then $b - a$ is

10. $\sqrt{11} + \sqrt{21} = \sqrt{\frac{a}{b}} + \sqrt{\frac{c}{d}}$, where a, b, c and d are natural numbers with $\gcd(a, b) = \gcd(c, d) = 1$.
Find $a + b + c + d$.

11. For $x \neq 0$ find the value of $\left(\frac{x^\ell}{x^m}\right)^{\ell^2+\ell m+m^2} \cdot \left(\frac{x^m}{x^n}\right)^{m^2+mn+n^2} \cdot \left(\frac{x^n}{x^\ell}\right)^{n^2+n\ell+\ell^2}$

12. For $a^x = (x + y + z)^y$, $a^y = (x + y + z)^z$, $a^z = (x + y + z)^x$, then find the values of x, y and z .
Where $a > 0$ and $a \neq 1$.

2. FACTORIZATIONS

2.1 $a^2 - b^2 = (a - b)(a + b)$

Illustration-14: $(3x - y)^2 - (2x - 3y)^2$

Solution : Use $a^2 - b^2 = (a - b)(a + b)$

$$(3x - y)^2 - (2x - 3y)^2 = (3x - y + 2x - 3y)(3x - y - 2x + 3y) = (5x - 4y)(x + 2y)$$

2.2 Factorising the Quadratic expression

Illustration-15: $x^2 + 6x - 187$

Solution : $x^2 + 6x - 187 = x^2 + 17x - 11x - 187$

$$= x(x + 17) - 11(x + 17)$$

$$= (x + 17)(x - 11)$$

2.3 Factorisation by converting the given expression into a perfect square.

Illustration-16: $9x^4 - 10x^2 + 1$

Solution :

$$\begin{aligned} 9x^4 - 10x^2 + 1 &= (3x^2)^2 - 2 \cdot 3x^2 + 1 - 4x^2 \\ &= (3x^2 - 1)^2 - (2x)^2 \\ &= (3x^2 - 1 - 2x)(3x^2 - 1 + 2x) \\ &= (x - 1)(3x + 1)(x + 1)(3x - 1) \end{aligned}$$

2.4 $a^3 \pm b^3 \equiv (a \pm b)(a^2 \mp ab + b^2)$

Illustration-17: $a^6 - b^6$

Solution :

$$\begin{aligned} a^6 - b^6 &= (a^2)^3 - (b^2)^3 \\ &= (a^2 - b^2)(a^4 + a^2b^2 + b^4) \\ &= (a - b)(a + b)(a^2 - ab + b^2)(a^2 + ab + b^2) \end{aligned}$$

2.5 Using Factor Theorem :

Illustration-18: $x^3 - 13x - 12$

Solution : As $x = -1$ makes given expression 0, $x + 1$ is a factor

$$\begin{array}{r} x+1 \overline{) x^3 - 13x - 12} \quad x^2 - x - 12 \\ \underline{x^3 + x^2} \\ -x^2 - 13x - 12 \\ \underline{-x^2 - x} \\ -12x - 12 \\ \underline{-12x - 12} \\ 0 \end{array}$$

$$\begin{aligned} \therefore x^3 - 13x - 12 &= (x + 1)(x^2 - x - 12) \\ &= (x + 1)(x - 4)(x + 3) \end{aligned}$$

2.6 $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$

Illustration-19: $8x^3 + y^3 + 27z^3 - 18xyz$

Solution :

$$\begin{aligned} 8x^3 + y^3 + 27z^3 - 18xyz &= (2x)^3 + (y)^3 + (3z)^3 - 3(2x)(y)(3z) \\ &= (2x + y + 3z)(4x^2 + y^2 + 9z^2 - 2xy - 6xz - 3yz) \end{aligned}$$

2.7 Cyclic Expression and its Factorization :

An expression is said to be cyclic with regard to the variables $x_1, x_2, \dots, x_n$ arranged in this order, when it is unchanged by changing x_1 into x_2 , x_2 into x_3 , x_3 into x_4 , ..., x_n into x_1 .

For three variables

$E(x, y, z)$ is cyclic if $E(x, y, z) = E(y, z, x) = E(z, x, y)$

Ex. 1. $x + y + z$ is a cyclic expression

2. $x^2 + y^2 + z^2 + xy + yz + zx$ is a cyclic expression

3. $E(x, y, z) = x(y - z) + y(z - x) + z(x - y)$ is a cyclic expression because

$$\Rightarrow E(y, z, x) = y(z - x) + z(x - y) + x(y - z)$$

$$\text{and } E(z, x, y) = z(x - y) + x(y - z) + y(z - x)$$

$$\Rightarrow E(x, y, z) = E(y, z, x) = E(z, x, y)$$

Theorem :

If $E(x, y, z)$ is a cyclic expression and $x - y$ is a factor of $E(x, y, z)$ then $y - z$ and $z - x$ are also factors of $E(x, y, z)$.

Illustration-20 : Factorize $x^2(y - z) + y^2(z - x) + z^2(x - y)$

Solution :

$$x^2(y - z) + x(z^2 - y^2) + yz(y - z)$$

$$= (y - z)(x^2 - x(z + y) + yz)$$

$$= (y - z)(x^2 - xz - xy + yz)$$

$$= (y - z)(x(x - z) - y(x - y))$$

$$= (y - z)(x - z)(x - y)$$

$$= -(x - y)(y - z)(z - x)$$

OR

$$\text{Let } E(x, y, z) = x^2(y - z) + y^2(z - x) + z^2(x - y)$$

$$\text{when } y = x, E(x, y, z) = x^2(x - z) + x^2(z - x) + z^2(x - x) = x^3 - x^2z + x^2z - x^3 = 0$$

$$\Rightarrow x - y \text{ is factor of } E(x, y, z)$$

So, $y - z$ and $z - x$ are also factors of $E(x, y, z)$

$$E(x, y, z) = A(x - y)(y - z)(z - x) \quad \dots(i)$$

Since degree of $E(x, y, z)$ is 3 so A is constant. We can find A by substituting values of x, y & z in (i)

$$\text{Let } x = 0, y = 1, z = 2$$

$$E(0, 1, 2) = A(-1)(-1)(2)$$

$$\Rightarrow 1(2 - 0) + 2^2(0 - 1) = 2A \Rightarrow A = -1$$

$$E(x, y, z) = -(x - y)(y - z)(z - x)$$

2.8 Important Algebraic Identities

- $xy + ay + bx + ab = (x + a)(y + b)$
- $x^2 + 2xy + y^2 = (x + y)^2$
- $x^2 + y^2 + z^2 + 2xy + 2yz + 2zx = (x + y + z)^2$
- $x^2 - y^2 = (x - y)(x + y)$
- $x^4 + x^2 + 1 = (x^2 + 1)^2 - x^2 = (x^2 + x + 1)(x^2 - x + 1)$
- $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$
- $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$
- $x^n - y^n = (x - y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \dots + y^{n-1}), n \in \mathbb{N}$
- $x^{2n+1} + y^{2n+1} = (x + y)(x^{2n} - x^{2n-1}y + x^{2n-2}y^2 - \dots + y^{2n}), n \in \mathbb{N}$

$$\bullet \quad x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$$

$$= \frac{1}{2}(x + y + z)\{(x - y)^2 + (y - z)^2 + (z - x)^2\}$$

$$\bullet \quad x^3 + y^3 + z^3 = 3xyz \text{ if } x + y + z = 0 \text{ or } x = y = z$$

$$\bullet \quad x^3 + y^3 + z^3 + 3(x + y)(y + z)(z + x) = (x + y + z)^3$$

Problem Solving Strategies :

- When facing a problem with gigantic numbers, try replacing them with smaller numbers and look for a pattern. You can often prove your pattern works and solve the problem by substituting variable expressions for the numbers.

Illustration-21 : Compute $\sqrt{2022 \times 2020 \times 2018 \times 2016 + 16}$ without using calculator.

Solution : Let $x = 2019$

$$\begin{aligned} & \sqrt{2022 \times 2020 \times 2018 \times 2016 + 16} \\ &= \sqrt{(x + 3)(x + 1)(x - 1)(x - 3) + 16} \\ &= \sqrt{(x^2 - 9)(x^2 - 1) + 16} \\ &= \sqrt{x^4 - 10x^2 + 25} = \sqrt{(x^2 - 5)^2} = x^2 - 5 \\ &= 2019^2 - 5 = (2000)^2 + (19)^2 + 2 \times 2000 \times 19 - 5 \\ &= 4000000 + 361 + 76000 - 5 \\ &= 4,076, 356 \end{aligned}$$

- Focusing on what makes a problem tricky helps identify what strategies might be best to solve the problem.
- If you have a problem that involves expressions of the form $a + b$ and $a - b$, where a and/or b involve square roots, consider finding a way to multiply the expressions to get rid of the square roots.

Illustration-22 : Suppose $x = z - \sqrt{z^2 - 5}$ and $5y = z + \sqrt{z^2 - 5}$.

Find x when $y = 2/3$.

Solution : $x = z - \sqrt{z^2 - 5}$ and $5y = z + \sqrt{z^2 - 5}$, multiplying both equations

$$\text{we get } 5xy = z^2 - (z^2 - 5) = 5$$

$$\Rightarrow xy = 1 \Rightarrow x = \frac{3}{2}.$$

- When making a substitution, take some time to look for the substitution that simplifies your work the most.
- If we have the product of two variables added to linear terms with both variables, such as $mn + 3m + 5n$, then there is a constant we can add that will allow us to factor. For example, adding 15 to $mn + 3m + 5n$ gives us $mn + 3m + 5n + 15 = (m + 5)(n + 3)$.

Illustration-23 : Find all integral solutions of x and y when $xy - y - 2x = 3$.

Solution :

$$xy - y - 2x = 3$$

$$\Rightarrow xy - y - 2x + 2 = 5$$

$$\Rightarrow y(x - 1) - 2(x - 1) = 5$$

$$\Rightarrow (x - 1)(y - 2) = 5$$

$x - 1$	$y - 2$	(x, y)
5	1	(6, 3)
1	5	(2, 7)
-5	-1	(-4, 1)
-1	-5	(0, -3)

All possible (x, y) and $(6, 3)$, $(2, 7)$, $(0, -3)$ and $(-4, 1)$

- Guessing has a long and glorious history in mathematics and science. It is often a very important first step in many discoveries. Don't be afraid to guess! But make sure you test your guesses – a guess itself is only a first step.

Illustration-24 : Find all pairs of positive integers m and n such that m^2 is 105 greater than n^2 .

Solution : Turning the words into math is easy :

$$m^2 = n^2 + 105.$$

$$\Rightarrow (m - n)(m + n) = 105.$$

$$(m - n)(m + n) = 1.105 = 3.35 = 5.21 = 7.15$$

Because m and n are positive, we know that $m - n$ is smaller than $m + n$, so we only have these four cases to consider :

$$m - n = 1$$

$$m - n = 3$$

$$m - n = 5$$

$$m - n = 7$$

$$m + n = 105$$

$$m + n = 35$$

$$m + n = 21$$

$$m + n = 15$$

Each of these systems of equations gives us a solution (m, n) . Adding the equations in the first case gives us $2m = 106$, so $m = 53$. Substitution then gives $n = 52$. Similarly, we can work through each of the other three cases to find the four solutions $(m, n) = (53, 52); (19, 16); (13, 8); (11, 4)$.

Illustration-25 : The number 7,999,999,999 has two prime factors. Find them.

Solution : Let $7,999,999,999 = 8,000,000,000 - 1$

$$= (2000)^3 - 1^3 = (2000 - 1)(2000^2 + 2000 + 1)$$

$$= (1999)(4002001)$$

According to question, 7,999,999,999 has two prime factors, they must be 1999 and 4002001.

Illustration-26 : Factor $x^4 + 4y^4$.

Solution :

$$x^4 + 4y^4 = (x^2)^2 + (2y^2)^2 - 2(x^2)(2y^2) + (2xy)^2$$

$$= (x^2 + 2y^2)^2 - (2xy)^2$$

$$= (x^2 + 2y^2 - 2xy)(x^2 + 2y^2 + 2xy)$$

Do yourself-2 :

1. Factorize following expressions

(i) $x^4 - y^4$ (ii) $9a^2 - (2x - y)^2$ (iii) $4x^2 - 9y^2 - 6x - 9y$

2. Factorize following expressions

(i) $8x^3 - 27y^3$ (ii) $8x^3 - 125y^3 + 2x - 5y$

3. Factorize following expressions

(i) $x^2 + 3x - 40$ (ii) $x^2 - 3x - 40$ (iii) $x^2 + 5x - 14$
 (iv) $x^2 - 3x - 4$ (v) $x^2 - 2x - 3$ (vi) $3x^2 - 10x + 8$
 (vii) $12x^2 + x - 35$ (viii) $3x^2 - 5x + 2$ (ix) $3x^2 - 7x + 4$
 (x) $7x^2 - 8x + 1$ (xi) $2x^2 - 17x + 26$ (xii) $3a^2 - 7a - 6$
 (xiii) $14a^2 + a - 3$

4. Factorize following expressions

- (i) $a^2 - 4a + 3 + 2b - b^2$ (ii) $x^4 + 324$ (iii) $x^4 - y^2 + 2x^2 + 1$
 (iv) $4a^4 - 5a^2 + 1$ (v) $4x^4 + 81$ (vi) $1 + x^4 + x^8$

5. Factorize following expressions

- (i) $x^3 - 6x^2 + 11x - 6$ (ii) $2x^3 + 9x^2 + 10x + 3$
 (iii) $2x^3 - 9x^2 + 13x - 6$ (iv) $x^6 - 7x^2 - 6$
 (v) $(x + y + z)^3 - x^3 - y^3 - z^3$

6. (i) Factorize the expressions $8a^6 + 5a^3 + 1$

- (ii) Show that $(x - y)^3 + (y - z)^3 + (z - x)^3 = 3(x - y)(y - z)(z - x)$.

7. Factorize following expressions

- (i) $(x + 1)(x + 2)(x + 3)(x + 4) - 15$
 (ii) $4x(2x + 3)(2x - 1)(x + 1) - 54$
 (iii) $(x - 3)(x + 2)(x + 3)(x + 8) + 56$

8. Factorize the following expressions :

- (i) $(x - y)(x - y - 1) - 20$ (ii) $(x + 2y)(x + 2y + 2) - 8$
 (iii) $3(4x + 5)^2 - 2(4x + 5) - 1$ (iv) $x^{\frac{2}{3}} + x^{\frac{1}{3}} - 2$
 (v) $x^6 - 7x^3 - 8$ (vi) $x^4 + 15x^2 - 16$
 (vii) $(x^2 - x - 3)(x^2 - x - 5) - 3$ (viii) $(x^2 + 5x + 6)(x^2 + 5x + 4) - 120$
 (ix) $(a^2 + 1)^2 + (a^2 + 5)^2 - 4(a^2 + 3)^2$
 (x) $(x - 1)^2 + (y - 1)^2 + (z - 1)^2 + 2(x - 1)(y - 1) + 2(y - 1)(z - 1) + 2(z - 1)(x - 1)$
 (xi) $2x^4 - x^3 - 6x^2 - x + 2$ (xii) $(x^4 + x^2 - 4)(x^4 + x^2 + 3) + 10$

2.9 Miscellaneous Algebraic Manipulations

Illustration-27 : Suppose $x + \frac{1}{x} = 5$ find (i) $x^2 + \frac{1}{x^2}$ (ii) $x^4 - \frac{1}{x^4}$

Solution : (i) Given : $x + \frac{1}{x} = 5$

$$\text{By squaring both sides } \left(x + \frac{1}{x}\right)^2 = (5)^2 \Rightarrow x^2 + \frac{1}{x^2} + 2 = 25 \Rightarrow x^2 + \frac{1}{x^2} = 23$$

$$(ii) \quad \left(x - \frac{1}{x}\right)^2 = \left(x + \frac{1}{x}\right)^2 - 4 = 21 \Rightarrow x - \frac{1}{x} = \pm\sqrt{21}$$

$$x^2 - \frac{1}{x^2} = \pm 5\sqrt{21}$$

$$\text{from (i) we have } x^2 + \frac{1}{x^2} = 23$$

$$\text{So } x^4 - \frac{1}{x^4} = \pm 23 \times 5\sqrt{21} = \pm 115\sqrt{21}$$

Illustration-28 : Simplify $\sqrt{6+\sqrt{11}} + \sqrt{6-\sqrt{11}}$

Solution : Let $x = \sqrt{6+\sqrt{11}} + \sqrt{6-\sqrt{11}}$

By squaring both sides

$$\begin{aligned} x^2 &= (\sqrt{6+\sqrt{11}})^2 + (\sqrt{6-\sqrt{11}})^2 + 2\sqrt{36-11} \\ &= 12 + 10 = 22 \end{aligned}$$

Since x is positive, so $x = \sqrt{22}$.

OR

$$\begin{aligned} x &= \sqrt{6+\sqrt{11}} + \sqrt{6-\sqrt{11}} = \frac{1}{\sqrt{2}} (\sqrt{12+2\sqrt{11}} + \sqrt{12-2\sqrt{11}}) \\ &= \frac{1}{\sqrt{2}} (\sqrt{(\sqrt{11}+1)^2} + \sqrt{(\sqrt{11}-1)^2}) = \frac{1}{\sqrt{2}} (\sqrt{11}+1 + \sqrt{11}-1) \end{aligned}$$

$$x = \sqrt{22}$$

Illustration-29 : If $\left(a + \frac{1}{a}\right)^2 = 3$, then $a^3 + \frac{1}{a^3}$ equals :

- (A) $6\sqrt{3}$ (B) $3\sqrt{3}$ (C) 0 (D) $6\sqrt{3}$

Solution : $a + \frac{1}{a} = \pm\sqrt{3}$

$$a^3 + \frac{1}{a^3} = \left(a + \frac{1}{a}\right)^3 - 3\left(a + \frac{1}{a}\right) = \pm 3\sqrt{3} \mp 3\sqrt{3} = 0$$

Illustration-30 : If $x = \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$ and $y = \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$, then find $x^3 + y^3$.

Solution :

$$x = \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}} = (\sqrt{3}-\sqrt{2})^2 = 5-2\sqrt{6}$$

$$y = 5+2\sqrt{6}$$

$$x^3 + y^3 = (x+y)(x^2 - xy + y^2)$$

$$= (x+y)[(x+y)^2 - 3xy] = 10 \times [100 - 3] = 970$$

Do yourself-3 :

1. Suppose $a + \frac{1}{a} = 3$:

(a) Find $a^2 + \frac{1}{a^2}$

(b) Find $a^4 + \frac{1}{a^4}$

(c) Find $a^3 + \frac{1}{a^3}$

2. If $x + \frac{1}{x} = 2$, then prove that : $x^2 + \frac{1}{x^2} = x^4 + \frac{1}{x^4} = x^8 + \frac{1}{x^8}$.

3. If $2x + 3y + 4z = 0$, then prove that $8x^3 + 27y^3 + 64z^3 = 72xyz$.

4. I'm thinking of two numbers. The sum of my numbers is 14 and the product of my numbers is 46. What is the sum of the squares of my numbers ?

5. Simplify $\sqrt{7-\sqrt{13}} - \sqrt{7+\sqrt{13}}$.

6. Simplify the expression $\sqrt[3]{2+\sqrt{5}} + \sqrt[3]{2-\sqrt{5}}$.

7. If x, y, z are all different real numbers, then prove that

$$\frac{1}{(x-y)^2} + \frac{1}{(y-z)^2} + \frac{1}{(z-x)^2} = \left(\frac{1}{x-y} + \frac{1}{y-z} + \frac{1}{z-x} \right)^2.$$

8. Solve for x :

(a) $\sqrt{2x+1} = \sqrt{x} + 1$

(b) $\sqrt{x^2-1} = x-3$

(c) $\sqrt{x+1} + 2\sqrt{2x-3} = -3$

3. POLYNOMIAL IN ONE VARIABLE

An algebraic expression of the form $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ is called a polynomial function in 'x' where $a_i (i = 0, 1, 2, \dots, n)$ is a constant which belongs to the set of real numbers and sometimes to the set of complex numbers and n is a whole number.

- a_i is the coefficient of x^i for $i = 1, 2, 3, \dots, n$ and a_0 is constant term of $p(x)$.
- If $a_n \neq 0$, then $a_n x^n$ is called leading term and a_n is called **leading co-efficient**.
- If $a_n = 1$, then polynomial is called **monic polynomial**.

- If $a_n \neq 0$, then **degree of the polynomial** is n .
- $f(x) = a_0$ is called **constant polynomial**. Its degree is 0, if $a_0 \neq 0$. If $a_0 = 0$ the polynomial $f(x)$ is called **ZERO polynomial**. Its degree is defined as $-\infty$ to preserve following two properties listed below. Some people prefer not to define degree of zero polynomial.

If $f(x)$ is a polynomial of degree n and $g(x)$ is a polynomial of degree m then

1. $f(x) \pm g(x)$ is a polynomial of degree $\leq \max\{n, m\}$
2. $f(x) \times g(x)$ is a polynomial of degree $m + n$.

3.1 Types of Polynomials (w.r.t. Degree)

Degree of the polynomial in one variable is the largest exponent of the variable. For example, the degree of the polynomial $3x^7 - 4x^6 + x + 9$ is 7 and the degree of the polynomial $5x^6 - 4x^2 - 6$ is 6.

Polynomials classified by degree

Degree	Name	General form	Example
undefined or $-\infty$	Zero polynomial	0	0
0	(Non-zero) constant polynomial	$a; (a \neq 0)$	4
1	Linear polynomial	$ax + b; (a \neq 0)$	$x + 1$
2	Quadratic polynomial	$ax^2 + bx + c; (a \neq 0)$	$x^2 + 1$
3	Cubic polynomial	$ax^3 + bx^2 + cx + d; (a \neq 0)$	$x^3 + 1$

Usually, a polynomial of degree n , for n greater than 3, is called a polynomial of degree n , although the phrases quartic polynomial for degree 4 and quintic polynomial for degree 5 are sometimes used.

Note :

Polynomials having only one term are called monomials. E.g. 2, $2x$, $7y^5$, $12t^7$ etc. Polynomials having exactly two dissimilar terms are called binomials. E.g. $p(x) = 2x + 1$, $r(y) = 2y^7 + 5y^6$ etc. Polynomials having exactly three distinct terms are called trinomials. E.g. $p(x) = 2x^2 + x + 6$, $q(y) = 9y^6 + 4y^2 + 1$ etc.

3.2 Division in Polynomials

Consider two polynomials $P(x)$ & $d(x)$ with $d(x)$ being not identically zero and degree of $d(x) \leq$ degree of $P(x)$ then there exists unique polynomials $Q(x)$ and $r(x)$ such that

$$P(x) = Q(x) \cdot d(x) + r(x)$$

Here $P(x)$ is called as dividend,

$d(x)$ is called as divisor,

$Q(x)$ is called as quotient,

& $r(x)$ is called as remainder with degree of $r(x) <$ degree of $d(x)$

Note : If $d(x)$ is a divisor of $P(x)$ then $kd(x)$ will also be a divisor of $P(x)$; $k \in \mathbb{R} - \{0\}$ and $d(-x)$ will be a divisor of $P(-x)$.

3.3 Remainder Theorem

Statement : Let $p(x)$ be a polynomial of degree ≥ 1 and 'a' is any real number. If $p(x)$ is divided by $(x - a)$, then the remainder is $p(a)$.

Illustration-31 : Let $p(x)$ be $x^3 - 7x^2 + 6x + 4$. Divide $p(x)$ with $(x - 6)$ and find the remainder

Solution : Put $x = 6$ in $p(x)$ i.e. $p(6)$ will be the remainder.

$\therefore$ required remainder be

$$p(6) = (6)^3 - 7 \cdot 6^2 + 6 \cdot 6 + 4 = 216 - 252 + 36 + 4 = 256 - 252 = 4$$

$$\begin{array}{r} x-6 \overline{) x^3 - 7x^2 + 6x + 4} \quad \left(x^2 - x \right. \\ \underline{-x^3 + 6x^2} \\ -x^2 + 6x + 4 \\ \underline{-x^2 + 6x} \\ + - \\ \text{Remainder} = 4 \end{array}$$

Thus, $p(a)$ is remainder on dividing $p(x)$ by $(x - a)$.

Remark : (i) $p(-a)$ is remainder on dividing $p(x)$ by $(x + a)$

$$[\because x + a = 0 \Rightarrow x = -a]$$

(ii) $p\left(\frac{b}{a}\right)$ is remainder on dividing $p(x)$ by $(ax - b)$

$$[\because ax - b = 0 \Rightarrow x = b/a]$$

(iii) $p\left(\frac{-b}{a}\right)$ is remainder on dividing $p(x)$ by $(ax + b)$

$$[\because ax + b = 0 \Rightarrow x = -b/a]$$

(iv) $p\left(\frac{b}{a}\right)$ is remainder on dividing $p(x)$ by $(b - ax)$

$$[\because b - ax = 0 \Rightarrow x = b/a]$$

Illustration-32 : Find the remainder when

$$x^3 - ax^2 + 6x - a \text{ is divided by } x - a$$

Solution :

$$\text{Let } p(x) = x^3 - ax^2 + 6x - a$$

$$p(a) = a^3 - a(a)^2 + 6(a) - a$$

$$= a^3 - a^3 + 6a - a = 5a$$

So, by the Remainder theorem, remainder = $5a$

3.4 Factor Theorem

Statement : Let $f(x)$ be a polynomial of degree ≥ 1 and a be any real constant such that $f(a) = 0$, then $(x - a)$ is a factor of $f(x)$. Conversely, if $(x - a)$ is a factor of $f(x)$, then $f(a) = 0$.

Proof : By Remainder theorem, if $f(x)$ is divided by $(x - a)$, the remainder will be $f(a)$. Let $q(x)$ be the quotient. Then, we can write,

$f(x) = (x - a) \times q(x) + f(a)$ ($\because$ Dividend = Divisor $\times$ Quotient + Remainder)

If $f(a) = 0$, then $f(x) = (x - a) \times q(x)$

Thus, $(x - a)$ is a factor of $f(x)$.

Converse Let $(x - a)$ is a factor of $f(x)$.

Then we have a polynomial $q(x)$ such that $f(x) = (x - a) \times q(x)$

Replacing x by a , we get $f(a) = 0$.

Hence, proved.

Illustration-33: Use the factor theorem to determine whether $(x - 1)$ is a factor of

$$f(x) = 2\sqrt{2}x^3 + 5\sqrt{2}x^2 - 7\sqrt{2}$$

Solution : By using factor theorem, $(x - 1)$ is a factor of $f(x)$, only when $f(1) = 0$

$$f(1) = 2\sqrt{2}(1)^3 + 5\sqrt{2}(1)^2 - 7\sqrt{2} = 2\sqrt{2} + 5\sqrt{2} - 7\sqrt{2} = 0$$

Hence, $(x - 1)$ is a factor of $f(x)$.

3.5 Fundamental Theorem of Algebra

Every polynomial function of degree ≥ 1 has atleast one zero in the complex numbers.

In other words, if we have

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0, a_i \in \text{complex number } \forall i = 0, 1, 2, \dots, n$$

with $n \geq 1$, then there exists atleast one $h \in \mathbb{C}$, such that

$$a_n h^n + a_{n-1} h^{n-1} + \dots + a_1 h + a_0 = 0.$$

From this, it is easy to deduce that a polynomial function of degree ' n ' (≥ 1) has exactly n zeroes.

i.e., $f(x) = a(x - r_1)(x - r_2)\dots(x - r_n)$

Illustration-34: Find the constants a, b, c such that $(2x^2 + 3x + 7)(ax^2 + bx + c) = 2x^4 + 11x^3 + 9x^2 + 13x - 35$

Solution : **Method : 1**

$$(2x^2 + 3x + 7)(ax^2 + bx + c) = 2x^4 + 11x^3 + 9x^2 + 13x - 35$$

By comparing coefficient of x^4 from both sides

$$2a = 2 \Rightarrow a = 1$$

By comparing coefficient of x^3 from both sides

$$2b + 3a = 11 \Rightarrow b = 4$$

By comparing coefficient of x^2 from both sides

$$2c + 3b + 7a = 9 \Rightarrow c = -5$$

By comparing coefficient of x from both sides

$$3c + 7b = 13 \quad (b \text{ \& } c \text{ satisfy})$$

By comparing constant

$$7c = -35 \Rightarrow c = -5$$

So

$$a = 1, b = 4, c = -5$$

Method : 2

Given that $(2x^2 + 3x + 7)(ax^2 + bx + c) = 2x^4 + 11x^3 + 9x^2 + 13x - 35$

$$\Rightarrow ax^2 + bx + c = \frac{2x^4 + 11x^3 + 9x^2 + 13x - 35}{2x^2 + 3x + 7}$$

$$= x^2 + 4x - 5$$

By comparing

$$a = 1, b = 4 \text{ and } c = -5.$$

Illustration 35 : Show that $(x - 3)$ is a factor of the polynomial $x^3 - 3x^2 + 4x - 12$.

Solution. Let $p(x) = x^3 - 3x^2 + 4x - 12$ be the given polynomial. By factor theorem, $(x - a)$ is a factor of a polynomial $p(x)$ iff $p(a) = 0$. Therefore, in order to prove that $x - 3$ is a factor of $p(x)$, it is sufficient to show that $p(3) = 0$.

$$\text{Now, } p(x) = x^3 - 3x^2 + 4x - 12$$

$$\Rightarrow p(3) = 3^3 - 3 \times 3^2 + 4 \times 3 - 12 = 27 - 27 + 12 - 12 = 0$$

Hence, $(x - 3)$ is a factor of $p(x) = x^3 - 3x^2 + 4x - 12$.

Illustration 36 : Without actual division prove that $2x^4 - 6x^3 + 3x^2 + 3x - 2$ is exactly divisible by $x^2 - 3x + 2$.

Solution. Let $f(x) = 2x^4 - 6x^3 + 3x^2 + 3x - 2$ and $g(x) = x^2 - 3x + 2$ be the given polynomials.

$$\text{Then } g(x) = x^2 - 3x + 2 = x^2 - 2x - x + 2 = x(x - 2) - 1(x - 2) = (x - 1)(x - 2)$$

In order to prove that $f(x)$ is exactly divisible by $g(x)$, it is sufficient to prove that $x - 1$ and $x - 2$ are factors of $f(x)$. For this it is sufficient to prove that $f(1) = 0$ and $f(2) = 0$.

$$\text{Now, } f(x) = 2x^4 - 6x^3 + 3x^2 + 3x - 2$$

$$\Rightarrow f(1) = 2 \times 1^4 - 6 \times 1^3 + 3 \times 1^2 + 3 \times 1 - 2 \Rightarrow f(1) = 0$$

$$\text{and, } f(2) = 2 \times 2^4 - 6 \times 2^3 + 3 \times 2^2 + 3 \times 2 - 2 \Rightarrow f(2) = 0$$

Hence, $(x - 1)$ and $(x - 2)$ are factors of $f(x)$.

$$\Rightarrow g(x) = (x - 1)(x - 2) \text{ is a factors of } f(x).$$

Hence, $f(x)$ is exactly divisible by $g(x)$.

Illustration 37 : The polynomials $P(x) = kx^3 + 3x^2 - 3$ and $Q(x) = 2x^3 - 5x + k$, when divided by $(x - 4)$ leave the same remainder. The value of k is

- (A) 2 (B) 1 (C) 0 (D) -1

Solution : $P(4) = 64k + 48 - 3 = 64k + 45$
 $Q(4) = 128 - 20 + k = k + 108$
 given $P(4) = Q(4)$
 $64k + 45 = k + 108$
 $\Rightarrow 63k = 63 \quad \Rightarrow k = 1 \quad \Rightarrow \text{Option (B) is correct}$

Illustration-38 : Let $P(x)$ be a polynomial such that when $P(x)$ is divided by $x - 19$, the remainder is 99, and when $P(x)$ is divided by $x - 99$, the remainder is 19. What is the remainder when $P(x)$ is divided by $(x - 19)(x - 99)$?

Solution : Because we are dividing by a quadratic, the degree of the remainder is not greater than 1. So, the remainder is $ax + b$, for some constants a and b . Therefore, we have

$$P(x) = (x - 19)(x - 99)Q(x) + ax + b,$$

where $Q(x)$ is the quotient when $P(x)$ is divided by $(x - 19)(x - 99)$. We eliminate the $Q(x)$ term by letting $x = 19$ or by letting $x = 99$. Doing each in turn gives us the system of equations

$$P(19) = 19a + b = 99,$$

$$P(99) = 99a + b = 19.$$

Solving this system of equations gives us $a = -1$ and $b = 118$.

So, the remainder is $-x + 118$.

Illustration-39 : The polynomial $f(x) = x^4 + ax^3 + bx^2 + cx + d$ has roots 1, 3, 5 and 7. Determine all the coefficients of $f(x)$.

Solution : By applying factor theorem

$$\begin{aligned} f(x) &= (x - 1)(x - 3)(x - 5)(x - 7) \\ &= (x^2 - 4x + 3)(x^2 - 12x + 35) \\ &= x^2(x^2 - 12x + 35) - 4x(x^2 - 12x + 35) + 3(x^2 - 12x + 35) \\ &= x^4 - 16x^3 + 86x^2 - 176x + 105 \end{aligned}$$

So $a = -16$, $b = 86$, $c = -176$ and $d = 105$

Illustration-40 : If $f(x)$ is monic polynomial of degree 6 such that $f(0) = 0$, $f(1) = -1$, $f(2) = -2$, $f(3) = -3$, $f(4) = -4$ and $f(5) = -5$, then find $f(x)$.

Solution : According to question

$$\begin{aligned} f(0) &= 0, f(1) = -1, f(2) = -2, \dots, f(5) = -5 \\ \Rightarrow f(x) + x &= 0 \text{ has the roots } x = 0, 1, 2, \dots, 5 \\ \Rightarrow f(x) + x &= x(x - 1)(x - 2)(x - 3)(x - 4)(x - 5) \text{ (By factor theorem)} \\ \Rightarrow f(x) &= x(x - 1)(x - 2)(x - 3)(x - 4)(x - 5) - x. \end{aligned}$$

Illustration-41 : If $P(x) = 2x^3 + ax^2 + bx + c$, where $a, b, c \in \mathbb{Z}$. If $P(\sqrt{3}) = 10 - 2\sqrt{3}$.

Find (i) $P(-\sqrt{3})$ (ii) $3a + c$

Solution :

$$P(x) = 2x^3 + ax^2 + bx + c$$

$$P(\sqrt{3}) = 10 - 2\sqrt{3}$$

$$\Rightarrow 6\sqrt{3} + 3a + b\sqrt{3} + c = 10 - 2\sqrt{3}$$

$$\Rightarrow (3a + c) + \sqrt{3}(6 + b) = 10 - 2\sqrt{3}$$

$$\Rightarrow 3a + c = 10 \text{ and } 6 + b = -2 \quad (a, b, c \in \mathbb{I})$$

$$3a + c = 10 \text{ and } b = -8$$

$$P(-\sqrt{3}) = -6\sqrt{3} + 3a - \sqrt{3}b + c$$

$$= -6\sqrt{3} + (3a + c) + 8\sqrt{3}$$

$$P(-\sqrt{3}) = 2\sqrt{3} + 10$$

Illustration-42 : Let $P(x) = x^4 + ax^3 + bx^2 + cx + d$, where a, b, c, d are constants. If $P(1) = 10$, $P(2) = 20$, $P(3) = 30$, then compute $P(4) + P(0)$.

Solution :

$$P(1) = 10, P(2) = 20 \text{ and } P(3) = 30$$

we can write these information as

$$P(x) = 10x \text{ for } x = 1, 2, 3$$

$$\Rightarrow P(x) - 10x = 0 \text{ has roots } x = 1, 2 \text{ and } 3$$

By factor theorem

$$P(x) - 10x = (x - \alpha)(x - 1)(x - 2)(x - 3)$$

$$\Rightarrow P(x) = (x - \alpha)(x - 1)(x - 2)(x - 3) + 10x$$

$$P(4) + P(0) = (4 - \alpha)(3)(2)(1) + 40 + (-\alpha)(-1)(-2)(-3) + 0$$

$$= 24 - 6\alpha + 40 + 6\alpha$$

$$= 64$$

Illustration-43 : Let a, b and c are roots of $2x^3 + x^2 + x + 1 = 0$

find (i) $a + b + c$ (ii) abc

Solution :

By factor theorem

$$2x^3 + x^2 + x + 1 = 2(x - a)(x - b)(x - c)$$

$$\Rightarrow 2x^3 + x^2 + x + 1 = 2(x^3 - (a + b + c)x^2 + (ab + bc + ca)x - abc)$$

$$\Rightarrow 2x^3 + x^2 + x + 1 = 2x^3 - 2(a + b + c)x^2 + 2(ab + bc + ca)x - 2abc$$

By comparing coefficient of x^2 and constant term, we have

$$-2(a + b + c) = 1 \text{ and } -2abc = 1$$

$$\Rightarrow a + b + c = -\frac{1}{2} \text{ and } abc = -\frac{1}{2}.$$

Do yourself-4 :

1. Determine the remainder when the polynomial $P(x) = x^4 - 3x^2 + 2x + 1$ is divided by $(x - 1)$.
2. Find the remainder when $f(x) = 3x^3 + 6x^2 - 4x - 5$ is divided by $(x + 3)$.
3. Determine the value of k for which $x^3 - 6x + k$ may be divisible by $(x - 2)$.
4. Find the value of a , if $(x - a)$ is a factor of $x^3 - a^2x + x + 2$.
5. Find remainder when $f(x) = x^5 - x^3 + 3x^2 + 3x + 1$ is divided by $(x^2 - 1)$.
6. Find the value of ℓ and m if $8x^3 + \ell x^2 - 27x + m$ is divisible by $2x^2 - x - 6$.
7. Find ℓ and m if $2x^3 - (2\ell + 1)x^2 + (\ell + m)x + m$ may be exactly divisible by $2x^2 - x - 3$.
8. $f(x)$ when divided by $x^2 - 3x + 2$ leaves the remainder $ax + b$. If $f(1) = 4$ and $f(2) = 7$, determine a and b .
9. A polynomial in x of the third degree which will vanish when $x = 1$ & $x = -2$ and will have the values 4 & 28 when $x = -1$ and $x = 2$ respectively. Find the polynomial.
10. If $f(x)$ is polynomial of degree 4 such that $f(1) = 1$, $f(2) = 2$, $f(3) = 3$, $f(4) = 4$ & $f(0) = 1$ find $f(5)$.
11. Suppose a and b are constants such that $(x^3 + bx^2 - 7x + 9)(x^2 + ax + 5) = x^5 + 13x^4 + 38x^3 - 22x^2 + 37x + 45 \forall x \in \mathbb{R}$. Find a and b .
12. If $P(x)$ is a cubic polynomial such that $P(1) = 1$; $P(2) = 2$; $P(3) = 3$ with leading coefficient 3 then find the value of $P(4)$.

4. EQUATIONS REDUCIBLE TO QUADRATIC EQUATIONS

There are certain equations which can be reduced to $ax^2 + bx + c = 0$ by some proper substitution.

4.1 $a(f(x))^2 + b(f(x)) + c = 0$, where $f(x)$ is expression of x

Method of solving : Put $f(x) = y$

Illustration-44 : (a) Solve $2^{x+3} + 2^{-x} - 6 = 0$ (b) Solve $\left(\frac{x}{x+1}\right)^2 + 6 - 5\left(\frac{x}{x+1}\right) = 0$

Solution : (a) Put $2^x = y$

$$8y + \frac{1}{y} - 6 = 0$$

$$8y^2 - 6y + 1 = 0$$

$$(4y - 1)(2y - 1) = 0$$

$$y = \frac{1}{4}, \frac{1}{2}$$

$$\therefore 2^x = 2^{-2} \text{ and } 2^x = 2^{-1}$$

$$\therefore x = -2, x = -1$$

(b) Put $\frac{x}{x+1} = y$
 $y^2 - 5y + 6 = 0$

$$(y - 2)(y - 3) = 0 \Rightarrow \frac{x}{x+1} = 2 \text{ and } \frac{x}{x+1} = 3$$

4.2 $(x - a)^4 + (x - b)^4 = c,$

Method of Solving : Put $\frac{x - a + x - b}{2} = t \Rightarrow x = \frac{(a + b)}{2} + t$

Illustration-45 : $(x - 1)^4 + (x - 7)^4 = 272$

Solution : Put $x = \frac{7+1}{2} + t$

$$\Rightarrow x = t + 4$$

$$\Rightarrow (t + 3)^4 + (t - 3)^4 = 272$$

$$\Rightarrow 2(t^4 + 6.9t^2 + 81) = 272$$

$$\Rightarrow t^4 + 54t^2 + 81 = 136$$

$$\Rightarrow t^4 + 54t^2 - 55 = 0$$

$$\Rightarrow (t^2 - 1)(t^2 + 55) = 0$$

$$\Rightarrow t^2 = 1$$

$$\Rightarrow t = \pm 1$$

$$\Rightarrow x = 5, 3$$

4.3 $ma^{2x} + n(ab)^x + rb^{2x} = 0 \quad (a \text{ \& } b > 0)$

Method of Solving : Divide the equation by b^{2x} and put $\left(\frac{a}{b}\right)^x = t$ for $t > 0$

Illustration-46 : $3^{2x+2} + 5.6^x - 4^{x+1} = 0$

Solution : $3^{2x+2} + 5.6^x - 4^{x+1} = 0$

$$\Rightarrow 9 \times (3)^{2x} + 5 \times (2 \times 3)^x - 4(2)^{2x} = 0$$

$$\Rightarrow 9\left(\frac{3}{2}\right)^{2x} + 5\left(\frac{3}{2}\right)^x - 4 = 0 \quad \dots(1)$$

$$\text{Let } \left(\frac{3}{2}\right)^x = t \text{ for } t > 0$$

Equation 1 becomes

$$9t^2 + 5t - 4 = 0$$

$$\Rightarrow t = \frac{-5 \pm \sqrt{25 + 144}}{18} = -1 \text{ or } \frac{4}{9}$$

$$t = -1 \text{ (rejected)}$$

$$t = \frac{4}{9} \Rightarrow \left(\frac{3}{2}\right)^x = \left(\frac{2}{3}\right)^2 = \left(\frac{3}{2}\right)^{-2}$$

$$\Rightarrow x = -2$$

Solution of the given equation is $x = -2$

4.4 $m \cdot a^{f(x)} + n \cdot b^{f(x)} + r = 0$, where $ab = 1$, a & $b > 0$ and $f(x)$ is expression of x

Method of Solving : put $a^{f(x)} = t$, then $b^{f(x)} = \frac{1}{t}$

Illustration-47 : Solve $(\sqrt{5+2\sqrt{6}})^x + (\sqrt{5-2\sqrt{6}})^x = 10$

Solution :

$$\text{Let } a = \sqrt{5+2\sqrt{6}} \text{ and } b = \sqrt{5-2\sqrt{6}}$$

$$ab = 1$$

$$\text{Let } a^x = t$$

Given equation become

$$t + \frac{1}{t} = 10 \Rightarrow t^2 - 10t + 1 = 0$$

$$\Rightarrow t = \frac{10 \pm \sqrt{96}}{2} = 5 \pm 2\sqrt{6}$$

$$\Rightarrow (5+2\sqrt{6})^{x/2} = (5+2\sqrt{6}) \Rightarrow \frac{x}{2} = 1 \Rightarrow x = 2$$

$$\text{or } (5+2\sqrt{6})^{x/2} = (5-2\sqrt{6}) = (5+2\sqrt{6})^{-1}$$

$$\Rightarrow \frac{x}{2} = -1 \Rightarrow x = -2$$

Solutions of the given equation is $x = 2$ or -2 .

4.5 $(x+a)(x+b)(x+c)(x+d) + e = 0$ when $b+c=a+d$

Illustration-48 : Solve $x(x+1)(x+2)(x+3) - 8 = 0$

Solution :

$$x(x+1)(x+2)(x+3) - 8 = 0$$

$$\Rightarrow x(x+3)(x+1)(x+2) - 8 = 0$$

$$\Rightarrow (x^2+3x)(x^2+3x+2) - 8 = 0$$

$$\Rightarrow (x^2+3x)^2 + 2(x^2+3x) - 8 = 0$$

$$\Rightarrow (x^2+3x)^2 + 4(x^2+3x) - 2(x^2+3x) - 8 = 0$$

$$\Rightarrow (x^2+3x)(x^2+3x+4) - 2(x^2+3x+4) = 0$$

$$\Rightarrow (x^2+3x-2)(x^2+3x+4) = 0$$

$$\Rightarrow (x^2+3x-2) = 0 \text{ or } (x^2+3x+4) = 0$$

$$\Rightarrow x = \frac{-3 \pm \sqrt{17}}{2} \text{ or } x = \frac{-3 \pm \sqrt{7}i}{2}$$

4.6 $(x+a)(x+b)(x+c)(x+d) + ex^2 = 0$, where $ad = bc$.

Method of solving : Divide given equation by x^2 and put $x + \frac{ad}{x} = t$

Illustration-49 : $(x+1)(x+2)(x+3)(x+6) = 3x^2$

Solution :

$$(x+1)(x+6)(x+2)(x+3) - 3x^2 = 0$$

$$\Rightarrow (x^2+7x+6)(x^2+5x+6) - 3x^2 = 0$$

$$\Rightarrow \left(x + \frac{6}{x} + 7\right) \left(x + \frac{6}{x} + 5\right) - 3 = 0$$

$$\text{Let } x + \frac{6}{x} = t$$

$$(t + 7)(t + 5) - 3 = 0$$

$$\Rightarrow t^2 + 12t + 32 = 0$$

$$t = -8 \text{ or } -4$$

$$\text{when } x + \frac{6}{x} = -8 \Rightarrow x^2 + 8x + 6 = 0 \Rightarrow x = -4 \pm \sqrt{10}$$

$$x + \frac{6}{x} = -4 \Rightarrow x^2 + 4x + 6 = 0 \Rightarrow x = -2 \pm \sqrt{2}i$$

Illustration-50 : Solve $(x^2 - 3x)(x^2 - 3x + 2) + 1 = 0$

Solution :

$$\text{Let } x^2 - 3x = y$$

$$\Rightarrow y(y + 2) + 1 = 0$$

$$\Rightarrow y^2 + 2y + 1 = 0$$

$$\Rightarrow (y + 1)^2 = 0$$

$$\Rightarrow y = -1$$

$$\text{Putting } y = -1 \text{ in } x^2 - 3x = y$$

$$\text{we have } x^2 - 3x = -1$$

$$\Rightarrow x^2 - 3x + 1 = 0$$

$$\Rightarrow x = \frac{3 \pm \sqrt{5}}{2}$$

4.7 $ax^4 + bx^3 + cx^2 + dx + e = 0$, where $a = e$ & $b = \pm d$

Method of solving : Divide given equation by x^2 and put $x + \frac{1}{x} = t$ or $x - \frac{1}{x} = t$ whichever is applicable.

Illustration-51 : Solve $x^4 - 2x^3 + 3x^2 - 2x + 1 = 0$

Solution :

$$x^4 - 2x^3 + 3x^2 - 2x + 1 = 0$$

By dividing x^2 both sides we have

$$x^2 - 2x + 3 - \frac{2}{x} + \frac{1}{x^2} = 0$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2\left(x + \frac{1}{x}\right) + 3 = 0$$

Let $x + \frac{1}{x} = t$

Above equation become

$$t^2 - 2 - 2t + 3 = 0$$

$$\Rightarrow t^2 - 2t + 1 = 0 \Rightarrow (t - 1)^2 = 0$$

$$\Rightarrow t = 1 \Rightarrow x + \frac{1}{x} = 1 \Rightarrow x^2 - x + 1 = 0 \Rightarrow x = \frac{1 \pm \sqrt{3}i}{2}$$

Roots are $\frac{1 \pm \sqrt{3}i}{2}$

4.8 By Guessing Rational Roots of Polynomial.

Illustration-52 : Solve : $x^4 + x^3 - 2x^2 - x + 1 = 0$

Solution :

Let $P(x) = x^4 + x^3 - 2x^2 - x + 1$

$P(1) = 0$ and $P(-1) = 0$

$\Rightarrow (x - 1)(x + 1)$ factor of $P(x)$

We can find other factor of $P(x)$ by dividing $x^2 - 1$ from $P(x)$.

$$P(x) = (x^2 - 1)(x^2 + x - 1) = 0$$

$$\Rightarrow x = \pm 1 \text{ or } x^2 + x - 1 = 0$$

$$\Rightarrow x = \pm 1 \text{ or } x = \frac{-1 \pm \sqrt{5}}{2}$$

solution of $P(x) = 0$ are $x = \pm 1$ or $\frac{-1 \pm \sqrt{5}}{2}$.

Do yourself-5 :

Solve the following equations for x :

1. $x^{2/3} + x^{1/3} - 2 = 0$

2. $x^{\frac{2}{5}} - 3x^{\frac{1}{5}} + 2 = 0$

3. $3x^4 - 8x^2 + 4 = 0$

4. $4^x - 3 \cdot 2^{x+3} + 128 = 0$

5. $x^2 + \frac{1}{x^2} - 5\left(x + \frac{1}{x}\right) + 8 = 0$

6. $2\left(x^2 + \frac{1}{x^2}\right) - 3\left(x - \frac{1}{x}\right) - 4 = 0$

7. $2^{2x+1} - 7 \times 10^x + 5^{2x+1} = 0$

8. $(x - 1)^4 + (x - 5)^4 = 82$

9. $(\sqrt{3} + \sqrt{2})^x + (\sqrt{3} - \sqrt{2})^x - 2\sqrt{3} = 0$ 10. $(3 + 2\sqrt{2})^{x/2} + (3 - 2\sqrt{2})^{x/2} - 34 = 0$
11. $x(x+1)(x+2)(x+3) = 24$ 12. $(x+1)(x+2)(x+3)(x+4) = 120$
13. $(x+1)(x+2)^2(x+4) - 2x^2 = 0$ 14. $x^4 - 2x^3 - 2x^2 + 2x + 1 = 0$
15. Match the values of x given in **Column-II** satisfying the exponential equation in Column-I (Do not verify). Remember that for $a > 0$, the term a^x is always greater than zero $\forall x \in \mathbb{R}$.

Column-I

Column-II

(A) $5^x - 24 = \frac{25}{5^x}$

(P) -3

(B) $(2^{x+1})(5^x) = 200$

(Q) -2

(C) $4^{2/x} - 5(4^{1/x}) + 4 = 0$

(R) -1

(D) $\frac{2^{x-1} \cdot 4^{x+1}}{8^{x-1}} = 16$

(S) 0

(E) $4^{x^2+2} - 9(2^{x^2+2}) + 8 = 0$

(T) 1

(F) $5^{2x} - 7^x - 5^{2x}(35) + 7^x(35) = 0$

(U) 2

(V) 3

(W) None

5. SYSTEM OF EQUATIONS

Observe the following Illustrations :

Illustrations-53 : If $x - y = 2$ and $xy = 24$, find the value of $\frac{1}{x} + \frac{1}{y}$.

Solution :

$$\begin{aligned} (x+y)^2 &= (x-y)^2 + 4xy = 4 + 4(24) \\ \Rightarrow (x+y)^2 &= 100 \\ \Rightarrow x+y &= 10, -10 \\ \therefore \frac{x+y}{xy} &= \frac{10}{24} = \frac{5}{12}; \frac{x+y}{xy} = -\frac{10}{24} = -\frac{5}{12} \end{aligned}$$

Illustrations-54 : If $2x - 3y - z = 0$ and $x + 3y - 14z = 0$, then find $\frac{x^2 + 3xy}{y^2 + z^2}$.

Solution :

$$\begin{aligned} \frac{2x}{z} - \frac{3y}{z} &= 1 \quad \& \quad \frac{x}{z} + \frac{3y}{z} = 14 \\ \text{Solving } \frac{x}{z} &= 5; \frac{y}{z} = 3 \\ \Rightarrow \frac{\left(\frac{x}{z}\right)^2 + \frac{3x}{z} \cdot \frac{y}{z}}{\left(\frac{y}{z}\right)^2 + 1} &= \frac{25 + 3(5)(3)}{(3)^2 + 1} = \frac{70}{10} = 7 \end{aligned}$$

Illustration-55 : Given $a + b = 20$ and $a^3 + b^3 = 800$, find $a^2 + b^2$.

Solution :

$$a + b = 20$$

$$\Rightarrow a^2 + b^2 + 2ab = 400 \quad \dots (1)$$

$$a^3 + b^3 = 800$$

$$\Rightarrow (a + b)(a^2 - ab + b^2) = 800$$

$$\Rightarrow a^2 + b^2 - ab = 40 \quad \dots (2)$$

By adding twice of second equation with first equation.

$$3(a^2 + b^2) = 480$$

$$\Rightarrow a^2 + b^2 = 160.$$

Illustrations-56 : $x(y + z) = 29$, $y(z + x) = 26$; $z(x + y) = 51$ find x, y, z .

Solution :

$$xy + zx = 29 \quad \dots (1)$$

$$yz + xy = 26 \quad \dots (2)$$

$$xz + yz = 51 \quad \dots (3)$$

$$\Rightarrow 2[xy + yz + zx] = 106$$

$$\Rightarrow xy + yz + zx = 53$$

$$\text{Now : } xy = 2, zx = 27; yz = 24$$

$$\Rightarrow x^2y^2z^2 = 24 \times 2 \times 27 = (36)^2$$

$$\Rightarrow xyz = \pm 36 \therefore (x, y, z) = \left(\frac{3}{2}, \frac{4}{3}, 18\right) \text{ or } \left(-\frac{3}{2}, -\frac{4}{3}, -18\right)$$

Illustrations-57 : If $x^3 + y^3 = 35$; and $x^2y + xy^2 = 30$, then find (x, y) .

Solution :

$$\frac{x^3 + y^3}{x^2y + xy^2} = \frac{(x + y)(x^2 - xy + y^2)}{xy(x + y)} = \frac{35}{30} = \frac{7}{6}$$

$$\Rightarrow 6x^2 - 13xy + 6y^2 = 0 \Rightarrow (3x - 2y)(2x - 3y) = 0 \Rightarrow 3x = 2y \text{ or } 2x = 3y$$

$$\text{Case -I : } 3x = 2y, x^3 + \left(\frac{3x}{2}\right)^3 = 35 \Rightarrow 35x^3 = 8 \times 35 \Rightarrow x = 2, y = 3$$

$$\text{Case -II : } 2x = 3y, x^3 + \left(\frac{2x}{3}\right)^3 = 35 \Rightarrow 35x^3 = 27 \times 35 \Rightarrow x = 3, y = 2$$

Illustrations-58 : If x, y and z are real numbers such that $x^2 + y^2 + 2z^2 - 4x + 2z - 2yz + 5 = 0$, find the value of $(x - y - z)$.

Solution :

$$x^2 + y^2 + 2z^2 - 4x + 2z - 2yz + 5 = 0$$

$$\Rightarrow x^2 - 4x + 4 + y^2 + z^2 - 2yz + z^2 + 2z + 1 = 0$$

$$\Rightarrow (x - 2)^2 + (y - z)^2 + (z + 1)^2 = 0$$

$$\Rightarrow x = 2, y = z, z = -1$$

$$\therefore x - y - z = 2 + 1 + 1 = 4$$

Illustrations-59 : $xy = 12$, $yz = 15$, $zx = 20$ find $x + y + z$; $x, y, z \in \mathbb{R}^+$

Solution :

$$(xyz)^2 = (3 \times 4 \times 5)^2$$

$$\Rightarrow xyz = 3 \times 4 \times 5$$

$$\Rightarrow z = 5, y = 3, x = 4$$

$$\Rightarrow x + y + z = 5 + 3 + 4 = 12$$

Do yourself-6 :

Solve the following systems of equations :

1.
$$\begin{cases} x + 2y = 7 \\ 3x + 5y = 18 \end{cases}$$

2.
$$\begin{cases} 3x - 2y = 13 \\ 4x + 5y = 25 \end{cases}$$

3.
$$\begin{cases} x^2 - y^2 = 16, \\ x + y = 8 \end{cases}$$

4.
$$\begin{cases} x - y = 1, \\ x^3 - y^3 = 7 \end{cases}$$

5. $x + y = 2$ and $x^3 + y^3 = 56$; $x, y \in \mathbb{R}$, then find x and y .

6.
$$\begin{cases} x^2 + y^2 + 6x + 2y = 0, \\ x + y + 8 = 0 \end{cases}$$

7.
$$\begin{cases} \frac{x}{y} - \frac{y}{x} = \frac{5}{6}, \\ x^2 - y^2 = 5 \end{cases}$$

8.
$$\begin{cases} \frac{x+y}{x-y} + \frac{x-y}{x+y} = \frac{13}{6}, \\ xy = 5 \end{cases}$$

9.
$$\begin{cases} \frac{1}{x+1} + \frac{1}{y} = \frac{1}{3}, \\ \frac{1}{(x+1)^2} - \frac{1}{y^2} = \frac{1}{4} \end{cases}$$

10.
$$\begin{cases} x^2 + y^2 = 25 - 2xy \\ y(x + y) = 10 \end{cases}$$

11.
$$\begin{cases} \frac{1}{y-1} - \frac{1}{y+1} = \frac{1}{x}, \\ y^2 - x - 5 = 0 \end{cases}$$

12.
$$\begin{cases} 2xy + y^2 - 4x - 3y + 2 = 0, \\ xy + 3y^2 - 2x - 14y + 16 = 0 \end{cases}$$

13.
$$\begin{cases} x^3 + y^3 = 7, \\ xy(x + y) = -2 \end{cases}$$

14.
$$\begin{cases} x^4 + y^4 = 82, \\ xy = 3 \end{cases}$$

6. INEQUALITIES

6.1 Basic Rules :

- If $a > b$ and $b > c$, then $a > c$.
- If $x > y$, then $x + c > y + c$ for any real number c . Additionally, if $a > b$, then $x + a > y + b$.
- If $x > y$ and $a > 0$, then $xa > ya$.
- If we multiply or divide an inequality by a negative number, we must reverse the sign.
For example, if $x > y$ and $a < 0$, then $xa < ya$.
- If $x > y > 0$ and $a > b > 0$, then $xa > yb$.

- If $x > y$ and x and y have the same sign (positive or negative), then $\frac{1}{x} < \frac{1}{y}$.

- If $x > y \geq 0$, then for any positive real number a , we have $x^a > y^a$.

In particular, if $0 < a < b$, then $\sqrt[n]{a} < \sqrt[n]{b}$ for all positive integral values of $n > 1$.

E.g. $\sqrt[4]{2} < \sqrt[4]{7}$, $\sqrt[3]{3} < \sqrt[3]{5}$, $\sqrt[5]{10} < \sqrt[5]{13}$ etc.

If two simple surds of different orders viz. $\sqrt[n]{a}$ and $\sqrt[m]{b}$ have to be compared, they have to be expressed as surds of the same order i.e. LCM of n and m .

Ex. compare $\sqrt[4]{6}$ and $\sqrt[3]{5}$,

we express both as the surds of 12th order.

$$\therefore \sqrt[4]{6} = \sqrt[12]{6^3} \text{ and } \sqrt[3]{5} = \sqrt[12]{5^4}. \text{ As } 6^3 < 5^4 \Rightarrow \sqrt[4]{6} < \sqrt[3]{5}$$

Illustration-60 : Solve :

$$(i) 7 - 3x \geq 8 + 2x \quad (ii) \frac{8-x}{7} \geq 4 \quad (iii) \frac{1}{x} \geq 5 \quad (iv) \frac{4}{x+1} \leq 2$$

Solution :

$$(i) 7 - 3x \geq 8 + 2x \Rightarrow 7 - 8 \geq 2x + 3x$$

$$\Rightarrow \frac{-1}{5} \geq x \Rightarrow x \in \left(-\infty, -\frac{1}{5}\right]$$

$$(ii) \frac{8-x}{7} \geq 4 \Rightarrow 8 - x \geq 28 \Rightarrow -x \geq 20$$

$$\Rightarrow x \leq -20 \Rightarrow x \in (-\infty, -20]$$

$$(iii) \frac{1}{x} \geq 5$$

$$\Rightarrow \infty > \frac{1}{x} \geq 5 \quad \Rightarrow 0 < x \leq \frac{1}{5} \quad \Rightarrow x \in \left(0, \frac{1}{5}\right]$$

$$(iv) \frac{4}{x+1} \leq 2$$

$$\Rightarrow -\infty < \frac{4}{x+1} < 0 \text{ or } 0 < \frac{4}{x+1} \leq 2 \quad \left(\frac{4}{x+1} = 0 \text{ is not possible} \right)$$

$$\Rightarrow 0 > \frac{x+1}{4} > -\infty \text{ or } \infty > \frac{x+1}{4} \geq \frac{1}{2}$$

$$\Rightarrow x+1 < 0 \text{ or } 2 \leq x+1$$

$$\Rightarrow x < -1 \text{ or } 1 \leq x \Rightarrow x \in (-\infty, -1) \cup [1, \infty)$$

Do yourself : 7

1. Determine which of the following statements is true or false. If it is false, provide an example that shows the statement is false. Assume a, b, c, x and y are real numbers.

(a) If $a \leq b$ and $b \leq c$, then $a < c$.

(b) If $a \geq b \geq a$, then $a = b$.

(c) If $a > b$, then $ac > bc$.

(d) If $a > b$ and $c \leq 0$, then $ac \leq bc$.

(e) If $x + a \geq y + a$, then $x \geq y$.

(f) If $x + a \geq y + b$, then $x \geq y$ and $a \geq b$.

2. Which fraction is larger ?

(a) $\frac{13}{17}$ or $\frac{17}{21}$

(b) $\frac{31}{35}$ or $\frac{37}{41}$

3. Which of the following numbers is largest : $2^{36}, 3^{30}, 5^{18}, 6^{12}, 7^8, 8^4$? (No calculators!)

4. What values of x satisfy the inequality, $7 - 3x < x - 1 \leq 2x + 9$?

5. Which of the following is greater $5 + \sqrt{3}, 3 + \sqrt{14}$? (Without calculating the values of $\sqrt{3}, \sqrt{14}$)

6.2 Trivial and Sum of squares (SOS) Inequality

The square of any real number is non-negative. So if x is real, then $x^2 \geq 0$. This is known as Trivial inequality. Equality holds only if $x = 0$.

Sum of squares (SOS) of real numbers is non negative. That is $\sum x_i^2 \geq 0$. This is known as SOS inequality.

Equality holds if $x_i = 0 \forall i$

Ex. $x, y, z \in \mathbb{R}$ and $x^2 + y^2 + z^2 = 0 \Rightarrow x = y = z = 0$.

Note :

- $f(x) = [g(x)]^{2n}$ where $n \in \mathbb{N} \Rightarrow f(x) \geq 0$
- $f(x) = [g(x)]^{1/2n}$, $n \in \mathbb{N}$, $g(x) \geq 0 \Rightarrow f(x) \geq 0$

Illustration-61 : If $a, b, c \in \mathbb{R}$ and $a^2 + b^2 + c^2 - ab - bc - ca = 0$, prove that $a = b = c$.

Solution :

$$\begin{aligned} a^2 + b^2 + c^2 - ab - bc - ca &= 0 \\ \Rightarrow 2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca &= 0 \\ \Rightarrow (a^2 - 2ab + b^2) + (b^2 - 2bc + c^2) + (c^2 - 2ac + a^2) &= 0 \\ \Rightarrow (a - b)^2 + (b - c)^2 + (c - a)^2 &= 0 \\ \Rightarrow a - b = b - c = c - a &= 0 \\ \Rightarrow a = b = c. \end{aligned}$$

Illustration-62 : For $x, y \in \mathbb{R}$, find the all possible values (range) of expression $4x^2 + 9y^2 - 12x + 6y$.

Solution :

$$\begin{aligned} \text{If } E(x, y) &= 4x^2 + 9y^2 - 12x + 6y \\ &= (2x)^2 - 2(2x) \times 3 + (3)^2 + (3y)^2 + 2(3y) + (1)^2 - 10 \\ &= (2x - 3)^2 + (3y + 1)^2 - 10 \\ \text{By sum of square (SOS), } (2x - 3)^2 + (3y + 1)^2 &\geq 0 \\ \Rightarrow E(x, y) &= (2x - 3)^2 + (3y + 1)^2 - 10 \geq 0 - 10 \\ \text{So Range of } E(x, y) &= [-10, \infty) \end{aligned}$$

Illustration-63 : Find all ordered pairs of real numbers (x, y) such that $(4x^2 + 4x + 3)(y^2 - 6y + 13) = 8$

Solution :

Rewriting the equation

$$((2x + 1)^2 + 2)((y - 3)^2 + 4) = 8$$

from the SOS, $(2x + 1)^2 \geq 0$ and $(y - 3)^2 \geq 0$,
so we have

$$\begin{aligned} (2x + 1)^2 + 2 &\geq 2 \text{ and } (y - 3)^2 + 4 \geq 4 \\ \Rightarrow ((2x + 1)^2 + 2)((y - 3)^2 + 4) &\geq 8 \\ \text{Equality holds only if} \\ (2x + 1)^2 = (y - 3)^2 = 0 &\Rightarrow (x, y) = \left(-\frac{1}{2}, 3\right) \end{aligned}$$

Illustration-64 : If $b^2 - 4ac < 0$ and $a > 0$, then show that $ax^2 + bx + c > 0 \forall x \in \mathbb{R}$

Solution :

$$\begin{aligned} ax^2 + bx + c &= a \left[x^2 + \frac{b}{a}x \right] + c \\ &= a \left[x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} \right] + c \\ &= a \left[\left(x + \frac{b}{2a} \right)^2 - \left(\frac{b^2 - 4ac}{4a} \right) \right] + c \\ &> 0 \quad \forall x \in \mathbb{R}. \text{ Hence proved} \end{aligned}$$

6.3 Mean :

In any collection of data a specific value between two extremes (minimum/maximum) is called a mean of the data.

For Two Variables :

Let x, y be two **positive** real numbers with $x \leq y$.

- The **Arithmetic Mean (A)** of x and y is $\frac{x+y}{2}$

$$\text{Observe } x \leq \frac{x+y}{2} \leq y$$

- The **Geometric Mean (G)** of x and y is $\sqrt{xy}$

$$\text{Observe } x \leq \sqrt{xy} \leq y$$

We have $x \leq G \leq A \leq y$

Equality holds only when $x = y$.

Proof For two positive numbers x and y , the Trivial inequality gives us $(\sqrt{x} - \sqrt{y})^2 \geq 0$

$$\Rightarrow x + y - 2\sqrt{xy} \geq 0$$

$$\Rightarrow \frac{x+y}{2} \geq \sqrt{xy} \Rightarrow A \geq G$$

Note :

- $x + \frac{1}{x} \geq 2 \quad \forall x > 0$ and $x + \frac{1}{x} \leq -2 \quad \forall x < 0$

Illustration : 65 Find all possible values (range) of the expression $x + \frac{4}{x}$, when $x \in \mathbb{R} - \{0\}$.

Sol.

$$\text{When } x > 0, \quad \frac{x + \frac{4}{x}}{2} \geq \sqrt{x \times \frac{4}{x}} = 2 \quad (\text{By } A \geq G)$$

$$\Rightarrow x + \frac{4}{x} \geq 4,$$

$$\text{So, when } x > 0, \text{ range} = [4, \infty) \quad \dots(1)$$

$$\text{for } x < 0, \quad \frac{x + \frac{4}{x}}{2} \leq -\sqrt{x \times \frac{4}{x}}$$

$$\Rightarrow x + \frac{4}{x} \leq -4 \quad \dots(2)$$

$$\text{From (1) and (2), range} = (-\infty, -4] \cup [4, \infty)$$

Do yourself : 8

1. Find the minimum value of $\frac{x^4 + 8}{x^2}$.
2. For $x < 0$, find the maximum value of $\frac{3x^2 + 12}{x}$.
3. If $a, b \in \mathbb{R}^+$, find minimum possible value of $(a+b)\left(\frac{1}{a} + \frac{1}{b}\right)$.

7. RATIO AND PROPORTION

If a and b be two quantities of the same kind, then their ratio is $a : b$; which may be denoted by the fraction $\frac{a}{b}$ (This may be an integer or fraction)

In the ratio $a : b$, a is the first term (Antecedent) and b is the second term (Consequent)

A ratio may be represented in a number of ways e.g. $\frac{a}{b} = \frac{ma}{mb} = \frac{na}{nb} = \dots$ where $m, n, \dots$ are non-zero numbers.

Let a, b, c, d be positive integers now to compare two ratios $a : b$ and $c : d$ we use following :

- $(a : b) > (c : d)$ if $ad > bc$
- $(a : b) = (c : d)$ if $ad = bc$
- $(a : b) < (c : d)$ if $ad < bc$

To compare two or more ratio, reduce them to common denominator.

Note :

- If $a > b > 0$ and $x > 0$, then $\frac{a}{b} > \frac{a+x}{b+x}$, e.g. $\frac{41}{40} > \frac{45}{44}$
- If $0 < a < b$ and $x > 0$, then $\frac{a}{b} < \frac{a+x}{b+x}$

Illustration-66 : What term must be added to each term of the ratio $5 : 37$ to make it equal to $1 : 3$?

Solution : Let x be added to each term of the ratio $5 : 37$.

$$\text{Then } \frac{x+5}{x+37} = \frac{1}{3} \Rightarrow 3x + 15 = x + 37 \text{ i.e. } x = 11$$

Illustration-67 : If $x : y = 3 : 4$; find the ratio of $7x - 4y : 3x + y$

Solution : $\frac{x}{y} = \frac{3}{4} \Rightarrow x = \frac{3}{4}y$

$$\text{Now } \frac{7x - 4y}{3x + y} = \frac{7 \cdot \frac{3}{4}y - 4y}{3 \cdot \frac{3}{4}y + y} \quad (\text{putting the value of } x)$$

$$= \frac{5y}{13y} = \frac{5}{13} \text{ i.e. } 5 : 13$$

7.1 Proportion :

When two ratios are equal, then the four quantities composing them are said to be proportional.

so, if $\frac{a}{b} = \frac{c}{d}$, then it is written as $a : b = c : d$ or $a : b :: c : d$

Where 'a' and 'd' are known as extremes and 'b' and 'c' are known as means.

(i) An important property of proportion : Product of extremes = product of means.

(ii) If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f}$ then each is equal to $\frac{a + c + e}{b + d + f}$

(iii) If $a : b = c : d$, then $b : a = d : c$ (Invertendo)

$$\text{i.e. } \frac{a}{b} = \frac{c}{d} \Rightarrow \frac{b}{a} = \frac{d}{c}$$

(iv) If $a : b = c : d$, then $a : c = b : d$ (Alternando)

$$\text{i.e. } \frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{c} = \frac{b}{d}$$

(v) If $a : b = c : d$, then $\frac{a + b}{b} = \frac{c + d}{d}$ (Componendo)

$$\text{i.e. } \frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{b} + 1 = \frac{c}{d} + 1$$

(vi) If $a : b = c : d$, then $\frac{a - b}{b} = \frac{c - d}{d}$ (Dividendo)

$$\text{i.e. } \frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{b} - 1 = \frac{c}{d} - 1$$

(vii) If $a : b = c : d$, then $\frac{a+b}{a-b} = \frac{c+d}{c-d}$ (Componendo and dividendo)

$$\text{i.e. } \frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{b} + 1 = \frac{c}{d} + 1 \Rightarrow \frac{a+b}{b} = \frac{c+d}{d} \quad \dots (1)$$

$$\frac{a}{b} - 1 = \frac{c}{d} - 1 \Rightarrow \frac{a-b}{b} = \frac{c-d}{d} \quad \dots (2)$$

Dividing equation (1) by (2) we obtain

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Illustration-68: If $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$ show that $\frac{x^3+a^3}{x^2+a^2} + \frac{y^3+b^3}{y^2+b^2} + \frac{z^3+c^3}{z^2+c^2} = \frac{(x+y+z)^3 + (a+b+c)^3}{(x+y+z)^2 + (a+b+c)^2}$

Solution :

$$\frac{x}{a} = \frac{y}{b} = \frac{z}{c} = k \text{ (constant)}$$

$$x = ak; y = bk; z = ck$$

substituting these values of x, y, z in the given expression, we obtain

$$\begin{aligned} \text{L.H.S.} &= \frac{a^3k^3 + a^3}{a^2k^2 + a^2} + \frac{b^3k^3 + b^3}{b^2k^2 + b^2} + \frac{c^3k^3 + c^3}{c^2k^2 + c^2} \\ &= \frac{a^3(k^3+1)}{a^2(k^2+1)} + \frac{b^3(k^3+1)}{b^2(k^2+1)} + \frac{c^3(k^3+1)}{c^2(k^2+1)} = \frac{(k^3+1)}{(k^2+1)} \cdot (a+b+c) \end{aligned}$$

$$\text{Now R.H.S} = \frac{(ak+bk+ck)^3 + (a+b+c)^3}{(ak+bk+ck)^2 + (a+b+c)^2}$$

$$= \frac{k^3(a+b+c)^3 + (a+b+c)^3}{k^2(a+b+c)^2 + (a+b+c)^2}$$

$$= \frac{(k^3+1)(a+b+c)^3}{(k^2+1)(a+b+c)^2} = \frac{(k^3+1)}{(k^2+1)} \cdot (a+b+c) = \text{L.H.S}$$

Illustration-69 : If $\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e}$, prove that

$$(ab + bc + cd + de)^2 = (a^2 + b^2 + c^2 + d^2) (b^2 + c^2 + d^2 + e^2)$$

Solution : $\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e}$, then we have

$$\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e} = \frac{\sqrt{(a^2 + b^2 + c^2 + d^2)}}{\sqrt{(b^2 + c^2 + d^2 + e^2)}} = k \text{ (say)}$$

$$\text{i.e. } a = bk \quad \therefore ab = b^2k$$

$$b = ck \quad \therefore bc = c^2k$$

$$c = dk \quad \therefore cd = d^2k$$

$$d = ek \quad \therefore de = e^2k$$

$$\text{so, } (a^2 + b^2 + c^2 + d^2) = k^2 (b^2 + c^2 + d^2 + e^2) \quad \dots (i)$$

$$\text{Now L.H.S.} = (ab + bc + cd + de)^2$$

$$= (kb^2 + kc^2 + kd^2 + ke^2)^2$$

$$= k^2(b^2 + c^2 + d^2 + e^2)^2$$

$$= k^2(b^2 + c^2 + d^2 + e^2) (b^2 + c^2 + d^2 + e^2)$$

$$= (a^2 + b^2 + c^2 + d^2) (b^2 + c^2 + d^2 + e^2) = \text{R.H.S} \quad (\text{by use of (i)})$$

Illustration-70 : Solve the equation $\frac{3x^4 + x^2 - 2x - 3}{3x^4 - x^2 + 2x + 3} = \frac{5x^4 + 2x^2 - 7x + 3}{5x^4 - 2x^2 + 7x - 3}$

Solution :
$$\frac{3x^4 + x^2 - 2x - 3}{3x^4 - x^2 + 2x + 3} = \frac{5x^4 + 2x^2 - 7x + 3}{5x^4 - 2x^2 + 7x - 3}$$

Applying componendo and dividendo, we have

$$\frac{3x^4}{x^2 - 2x - 3} = \frac{5x^4}{2x^2 - 7x + 3}$$

$$\text{or } 3x^4 (2x^2 - 7x + 3) - 5x^4 (x^2 - 2x - 3) = 0$$

$$\text{or } x^4 [6x^2 - 21x + 9 - 5x^2 + 10x + 15] = 0$$

$$\text{or } x^4 (x^2 - 11x + 24) = 0$$

$$\therefore x = 0 \text{ or } x^2 - 11x + 24 = 0$$

$$\therefore x = 0 \text{ or } (x - 8) (x - 3) = 0$$

$$\therefore x = 0, 8, 3$$

Do yourself : 9

1. If $\frac{a}{b} = \frac{2}{3}$ and $\frac{b}{c} = \frac{4}{5}$, then find value of $\frac{a+b}{b+c}$
2. If $\frac{a}{b} = \frac{3}{5}$ and $\frac{b}{c} = \frac{7}{13}$, then find the value of $a : b : c$
3. If sum of two numbers is s and their quotient is $\frac{p}{q}$. Find number.
4. If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f}$, then find the value of $\frac{2a^4b^2 + 3a^2c^2 - 5e^4f}{2b^6 + 3b^2d^2 - 5f^5}$ in terms of a and b .
5. If $x : a = y : b = z : c$, then show that $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

8. SIGN-SCHEME (WAVY CURVE) METHOD

Given $f(x)$ and $g(x)$ are polynomials.

To solve the inequalities of the type $\frac{f(x)}{g(x)} * 0$, where '*' can be $>$, $\geq$, $<$ or $\leq$, we take the following steps :

- (i) Find all the roots of $f(x) = 0$ and $g(x) = 0$
- (ii) Write all these roots on the real line in increasing order of values.
- (iii) Check the sign of the expression $\frac{f(x)}{g(x)}$ at some x greater than the largest root. If it is positive, put + sign in rightmost interval. In case of negative, put -ve sign in rightmost interval and while moving from right to left change sign in accordance with step (iv).
- (iv) If a root occurs even number of times, then sign of expression will be same on both sides of the root and if a root occurs odd number of times, then sign of the expression will be different on both sides of the root.
- (v) Write the answer according to need of the question.

Note :

- We don't give equality sign on ' $\pm\infty$ ' in the solution as they are two improper points of number line.
- We can't take zeroes of denominator in the final answer as at these points expression is not defined (because division by '0' is not defined).
- In case of ≥ 0 or ≤ 0 , zeroes of numerator will be part of the answer provided they are not appearing in denominator also.
- Do not cross multiply the terms in the inequalities

Illustration-71 : Find the solution of $-x^2 + 6x + 7 \geq 0$

Solution :

$$-x^2 + 6x + 7 \geq 0$$

$$\Rightarrow x^2 - 6x - 7 \leq 0$$

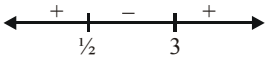
$$\Rightarrow (x + 1)(x - 7) \leq 0$$


$$\Rightarrow x \in [-1, 7]$$

Illustration-72 : Find the solution of $2x^2 - 7x + 3 \geq 0$

Solution :

$$2x^2 - 7x + 3 \geq 0$$

$$\Rightarrow (2x - 1)(x - 3) \geq 0$$


$$x \in \left(-\infty, \frac{1}{2}\right] \cup [3, \infty)$$

Illustration-73 : Solve $2x^4 > 3x^3 + 9x^2$

Solution :

$$2x^4 - 3x^3 - 9x^2 > 0$$

$$x^2(2x^2 - 3x - 9) > 0$$

$$x^2(2x + 3)(x - 3) > 0$$


$$x \in \left(-\infty, -\frac{3}{2}\right) \cup (3, \infty)$$

Illustration-74 : Find the solution of the inequalities $\frac{(x-1)(x-2)}{(x-3)} \geq 0$

Solution :

$$x - 1 = 0, x - 2 = 0, x - 3 = 0$$

$$\Rightarrow x = 1, 2, 3$$

Since $x - 3 \neq 0, x \neq 3$

so, $x \in [1, 2] \cup (3, \infty)$

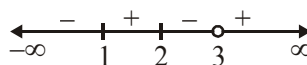
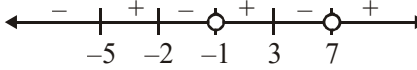


Illustration-75: If $f(x) = \frac{x^3 + 4x^2 - 11x - 30}{x^2 - 6x - 7}$ then find x such that

- (i) $f(x) > 0$ (ii) $f(x) < 0$.

Solution : Given $f(x) = \frac{(x-3)(x+2)(x+5)}{(x+1)(x-7)}$ 

- (i) $f(x) > 0 \Rightarrow x \in (-5, -2) \cup (-1, 3) \cup (7, \infty)$
(ii) $f(x) < 0 \Rightarrow x \in (-\infty, -5) \cup (-2, -1) \cup (3, 7)$

Illustration 76: Solve for real x : $\frac{1}{x+1} + \frac{2}{x+2} > \frac{3}{x+3}$

Solution : $\frac{3x+4}{(x+1)(x+2)} > \frac{3}{x+3}$

$$\Rightarrow \frac{3x+4}{(x+1)(x+2)} - \frac{3}{x+3} > 0 \Rightarrow \frac{4x+6}{(x+1)(x+2)(x+3)} > 0$$

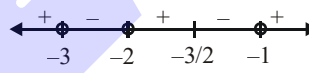
so, $x \in (-\infty, -3) \cup \left(-2, -\frac{3}{2}\right) \cup (-1, \infty)$ 

Illustration 77: Let $f(x) = \frac{(x-1)^3(x+2)^4(x-3)^5(x+6)}{x^2(x-7)^3}$. Solve the following inequality

- (i) $f(x) > 0$ (ii) $f(x) \geq 0$ (iii) $f(x) < 0$ (iv) $f(x) \leq 0$

Solution : We mark on the number line zeroes of numerator of expression : 1, -2, 3 and -6 (with black circles) and the zeroes of denominator 0 and 7 (with white circles), isolate the double points : -2 and 0 and draw the wavy curve :



From graph, we get

- (i) $x \in (-\infty, -6) \cup (1, 3) \cup (7, \infty)$
(ii) $x \in (-\infty, -6] \cup \{-2\} \cup [1, 3] \cup (7, \infty)$
(iii) $x \in (-6, -2) \cup (-2, 0) \cup (0, 1) \cup (3, 7)$
(iv) $x \in [-6, 0) \cup (0, 1] \cup [3, 7)$

Do yourself-10 :

Solve following inequalities over the set of real numbers :

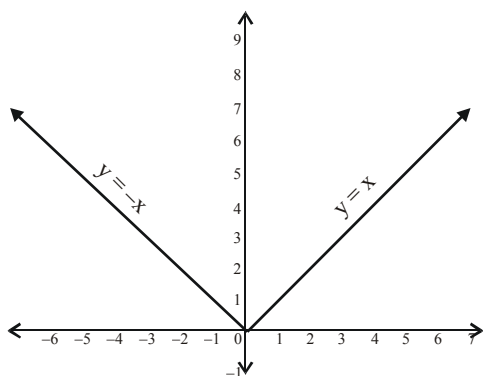
1. $(x-1)^2(x+1)^3(x-4) < 0$
 2. $\frac{6x-5}{4x+1} < 0$
 3. $\frac{(x-1)(x+2)^2}{-1-x} < 0$
 4. $\frac{(2x-1)(x-1)^2(x-2)^3}{(x-4)^4} > 0$
 5. $\frac{(x-1)^2(x+1)^3}{x^4(x-2)} \leq 0$
 6. $\frac{(x-2)^2(1-x)(x-3)^3(x-4)^2}{(x+1)} \leq 0$
 7. $\frac{x^2+4x+4}{2x^2-x-1} < 0$
 8. $\frac{x^3(x-2)(5-x)}{(x^2-4)(x+1)} > 0$
 9. $\frac{(2-x^2)(x-3)^3}{(x+1)(x^2-3x-4)} \geq 0$
 10. $x^4 - 5x^2 + 4 < 0$
 11. $\frac{15-4x}{x^2-x-12} < 4$
 12. $\frac{x^2+1}{4x-3} > 2$
 13. $\frac{1}{x-2} + \frac{1}{x-1} > \frac{1}{x}$
 14. $(x-2)(x+3) \geq 0$
 15. $\frac{x}{x+1} > 2$
 16. $\frac{3x-1}{4x+1} \leq 0$
- 17. Solve following Inequalities over the set of real numbers -**
- (i) $x^2 + 2x - 3 \leq 0$
 - (ii) $x^2 + 6x - 7 \leq 0$
 - (iii) $x^4 - 2x^2 - 63 \leq 0$
 - (iv) $\frac{x+1}{(x-1)^2} < 1$
 - (v) $\frac{x^2-7x+12}{2x^2+4x+5} > 0$
 - (vi) $\frac{(x-1)(x+2)^2}{-1-x} < 0$
 - (vii) $\frac{x^4+x^2+1}{x^2-4x-5} < 0$
 - (viii) $\frac{x+7}{x-5} + \frac{3x+1}{2} \geq 0$
 - (ix) $\frac{1}{x+2} < \frac{3}{x-3}$
 - (x) $\frac{14x}{x+1} - \frac{9x-30}{x-4} < 0$
 - (xi) $\frac{x^2+2}{x^2-1} < -2$
 - (xii) $\frac{5-4x}{3x^2-x-4} < 4$
 - (xiii) $\frac{(x+2)(x^2-2x+1)}{4+3x-x^2} \geq 0$
 - (xiv) $\frac{x^4-3x^3+2x^2}{x^2-x-30} > 0$
 - (xv) $\frac{2x}{x^2-9} \leq \frac{1}{x+2}$
 - (xvi) $\frac{20}{(x-3)(x-4)} + \frac{10}{x-4} + 1 > 0$

9. MODULUS AND MODULUS EQUATIONS

For any real number x , modulus or absolute value of x is denoted by $|x|$ and is defined as

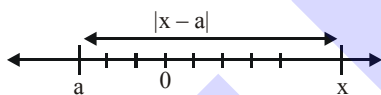
$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

- Graph of $y = |x|$



Note :

- $|x| = |-x| \geq 0$
- Geometrically $|x|$ is distance of real number x from zero along the real number line
- More generally $|x - a|$ is distance between ' x ' and ' a ' on the number line.



- $|x| = \sqrt{x^2}$
- $|xy| = |x| |y|$

Illustration-78 : Sketch the graph of following equation and also find all possible values (Range) of y

(i) $y = |x| + x$

(ii) $y = |x - 2| + x + 1$

(iii) $y = |x - 3| - |x|$

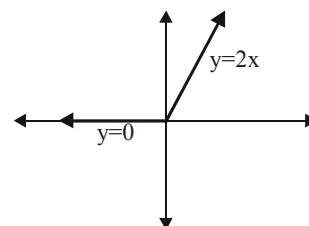
(iv) $y = 2|x - 2| - |x + 1|$

Solution :

$$(i) \quad y = |x| + x = \begin{cases} x + x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

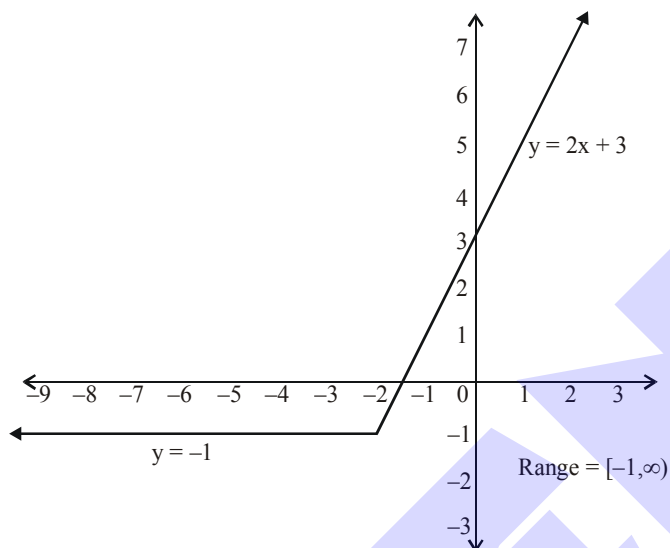
$$= \begin{cases} 2x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

From graph we can find all possible values (range) of y which is $[0, \infty)$



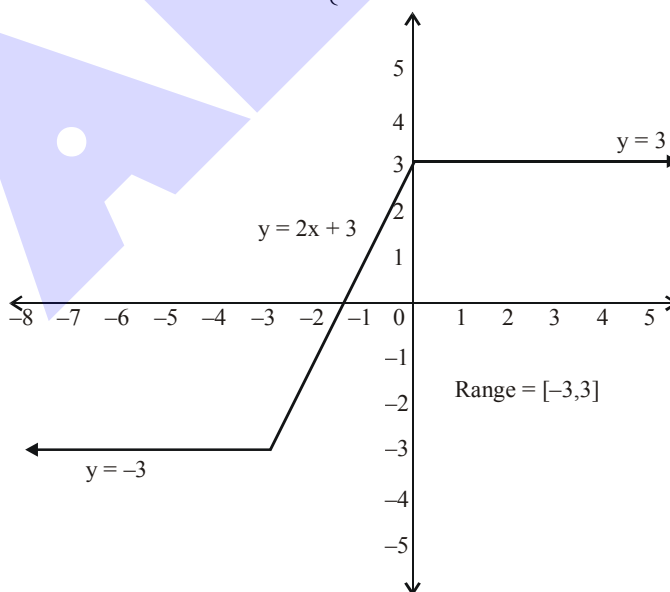
$$(ii) \quad y = |x + 2| + x + 1 = \begin{cases} x + 2 + x + 1 & , \quad x \geq -2 \\ -x - 2 + x + 1 & , \quad x < -2 \end{cases}$$

$$= \begin{cases} 2x + 3 & , \quad x \geq -2 \\ -1 & , \quad x < -2 \end{cases}$$



$$(iii) \quad y = |x + 3| - |x| = \begin{cases} x + 3 - x & , \quad x \geq 0 \\ x + 3 - (-x) & , \quad x \in [-3, 0) \\ -x - 3 - (-x) & , \quad x < -3 \end{cases}$$

$$= \begin{cases} 3 & , \quad x \geq 0 \\ 2x + 3 & , \quad x \in [-3, 0) \\ -3 & , \quad x < -3 \end{cases}$$



$$(iv) \quad y = 2|x - 2| - |x + 1| = \begin{cases} 2(x - 2) - (x + 1) & , \quad x \geq 2 \\ 2(-x + 2) - (x + 1) & , \quad x \in [-1, 2] \\ 2(-x + 2) + (x + 1) & , \quad x < -1 \end{cases}$$

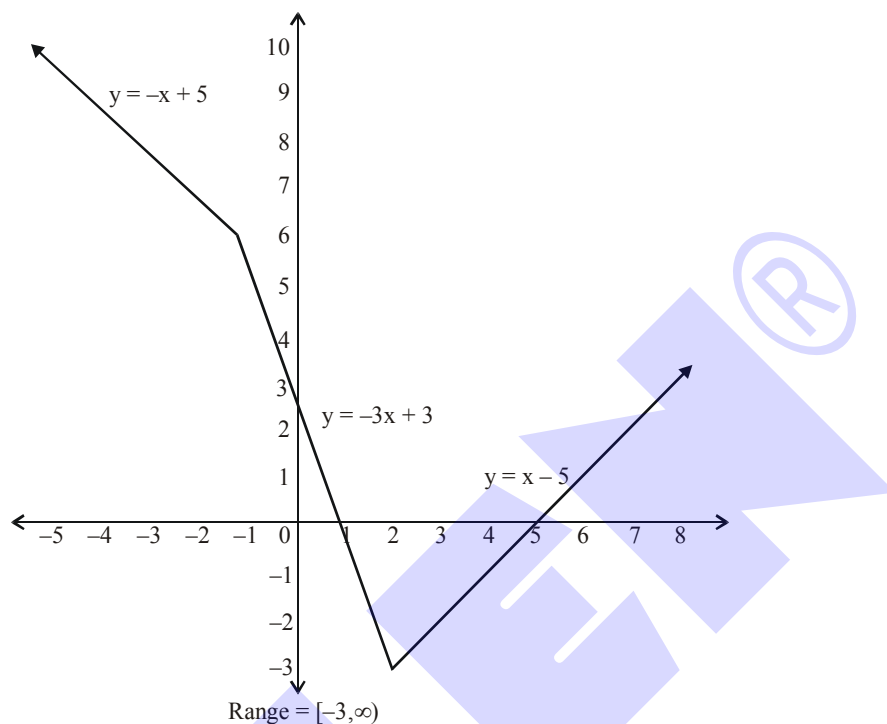


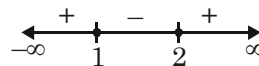
Illustration-79 : If $|x - 1||x - 2| = -(x^2 - 3x + 2)$, then find the interval in which x lies ?

Solution :

$$|(x - 1)(x - 2)| = -(x - 2)(x - 1)$$

$$\Rightarrow (x - 1)(x - 2) \leq 0$$

$$\Rightarrow 1 \leq x \leq 2$$



Do yourself-11

(1) Sketch the graph of following

(i) $y = |x - 2|$

(ii) $y = |x| - 2$

(iii) $y = 5 - |x|$

Solve for x

(2) $|2x + 5| = 2$

(3) $|2x - 5| = 7$

(4) $|x - 3| = -1$

(5) $|2x - 3| + 4 = 2$

(6) $\left| \frac{3x + 4}{3} \right| = 7$

(7) $|x^2 - 3x + 2| = 2$

(8) $|2x - 3| = |3x + 5|$

(9) $2|x + 3| = 3|x - 4|$

(10) $|x^2 + x + 1| = |x^2 + x + 2|$

(11) $|x^2 - 4x + 3| = |x^2 - 5x + 4|$

(12) $|x - 6| + |x - 3| = 1$

(13) $|2x - 1| + |2x + 3| = 6$

(14) $|3x + 5| + |4x + 7| = 12$

(15) $|x| + |x + 1| + |x + 2| = 3$

(16) $|2x - 3| + |2x + 1| + |2x + 5| = 12$

(17) $|x| - 2|x + 1| + 3|x + 2| = 0$

(18) $|x| - x = 0$

(19) $|x^2 + 3x + 2| + x + 1 = 0$

(20) $x^2 + 3|x| + 2 = 0$

(21) $|x^2 + 1| - x^2 - 1 = 0$

Useful Mathematical Symbols for Reference

Symbol	How it is read
$\forall$	For all...
$\exists$	There exists...
$\wedge$	and
$\vee$	or
$<$	is less than
$>$	is greater than
$\leq$	is less than or equal to
$\geq$	is greater than or equal to
$\nless$	is not less than
$\ngtr$	is not greater than
$\in$	belongs to
$\notin$	does not belong to
$, :, \text{ s.t.}$	such that
$\Rightarrow$	implies (If... then...)
$\Leftrightarrow$	implies and implied by (if and only if / iff)
$!$	factorial
$\sqrt{\quad}$	The square root of
$\sqrt[n]{\quad}$	n^{th} root, $n \in \mathbb{N}$, $n \geq 2$
Σ	The summation of
Π	The product of

Greek Letters (Capital, Small) and there pronunciations

Symbol (Capital, Small)	How it is read	Symbol (Capital, Small)	How it is read
A, α	alpha	N, ν	nu
B, β	beta	Ξ, ξ	xi
Γ, γ	gamma	O, o	omicron
Δ, δ	delta	Π, π	pi
E, ϵ	epsilon	P, ρ	rho
Z, ζ	zeta	Σ, σ	sigma
H, η	eta	T, τ	tau
Θ, θ	theta	Y, υ	upsilon
I, ι	iota	Φ, ϕ	phi
K, κ	kappa	X, χ	chi
Λ, λ	lambda	Ψ, ψ	psi
M, μ	mu	Ω, ω	omega